US012418159B2

(12) United States Patent
Koerner et al.

(10) Patent No.: US 12,418,159 B2
(45) Date of Patent: Sep. 16, 2025

(54) **METHOD OF FABRICATING A VCSEL DEVICE AND VCSEL DEVICE**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 806 days.

(21) Appl. No.: **17/718,394**

(22) Filed: **Apr. 12, 2022**

(65) **Prior Publication Data**

US 2022/0247152 A1 Aug. 4, 2022

Related U.S. Application Data

(63) Continuation of application No. PCT/EP2020/078400, filed on Oct. 9, 2020.

(30) **Foreign Application Priority Data**

Oct. 16, 2019 (EP) 19203671

(51) **Int. Cl.**
H01S 5/00 (2006.01)
H01S 5/02345 (2021.01)
(Continued)

(52) **U.S. Cl.**
CPC ***H01S 5/18322*** (2013.01); ***H01S 5/02345*** (2021.01); ***H01S 5/02476*** (2013.01);
(Continued)

(58) **Field of Classification Search**
CPC H01S 5/18322; H01S 5/02345; H01S 5/02476; H01S 5/04257; H01S 5/183;
(Continued)

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Primary Examiner — Kinam Park

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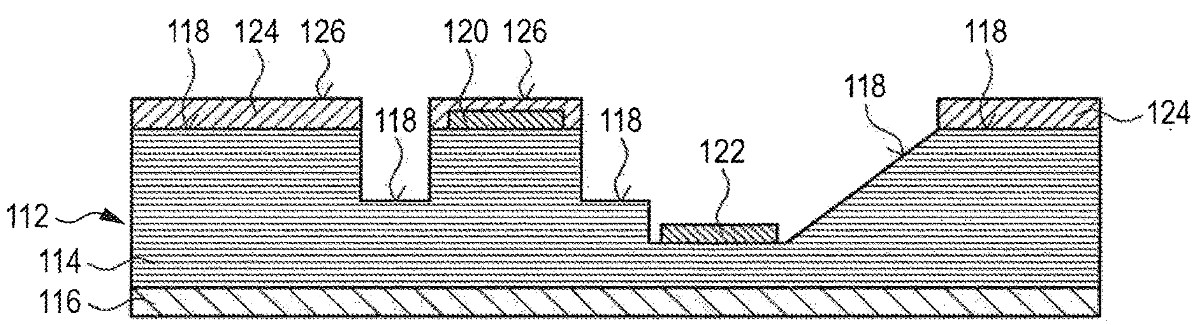
(57) **ABSTRACT**

A method of fabricating a Vertical Cavity Surface Emitting Laser(VCSEL) device includes providing a first structure comprising a VCSEL layer structure on a wafer. The first structure has a non-planar first structure top surface with varying height levels and includes one or more electrical contact areas. The method further includes applying one or more layers of cover material on the non-planar first structure top surface with a thickness such that a lowest height level of a cover material top surface is equal to or above the highest height level of the non-planar first structure top surface, to obtain a second structure having a second structure top surface, planarizing the second structure top surface, and producing one or more first electrical vias from the second structure top surface through the one or more layers of cover material for electrical connection to the one or more electrical contact areas.

12 Claims, 14 Drawing Sheets



(51) **Int. Cl.**
H01S 5/024 (2006.01)
H01S 5/042 (2006.01)
H01S 5/183 (2006.01)
H01S 5/42 (2006.01)

(52) **U.S. Cl.**
CPC ***H01S 5/04257*** (2019.08); ***H01S 5/183*** (2013.01); ***H01S 5/18305*** (2013.01); ***H01S 5/18347*** (2013.01); ***H01S 5/18361*** (2013.01); ***H01S 5/423*** (2013.01); *H01S 5/02469* (2013.01); *H01S 5/426* (2013.01)

(58) **Field of Classification Search**
CPC H01S 5/18305; H01S 5/18347; H01S 5/18361; H01S 5/423; H01S 5/02469; H01S 5/426; H01S 5/028
See application file for complete search history.

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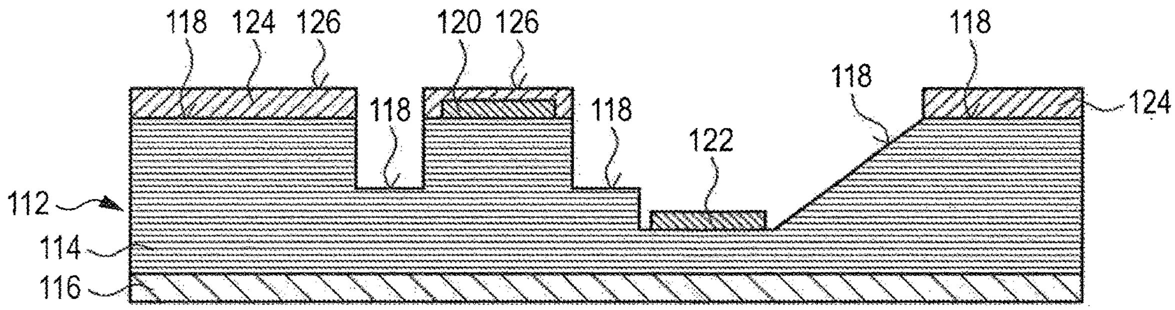


Fig. 1A

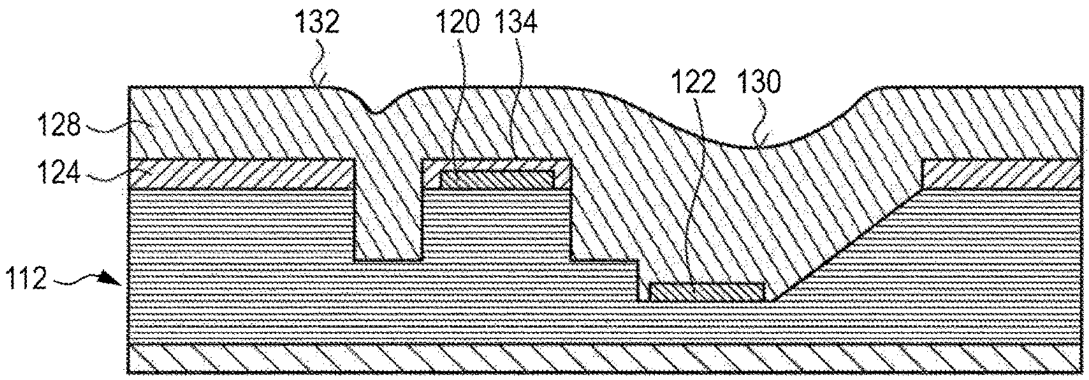


Fig. 1B

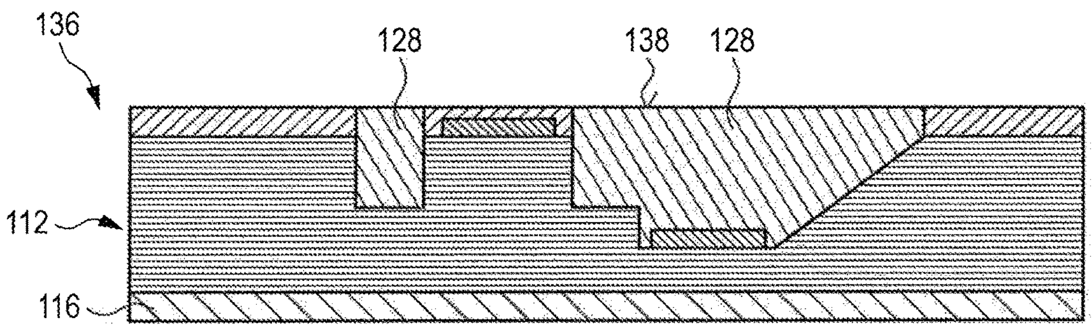


Fig.1C

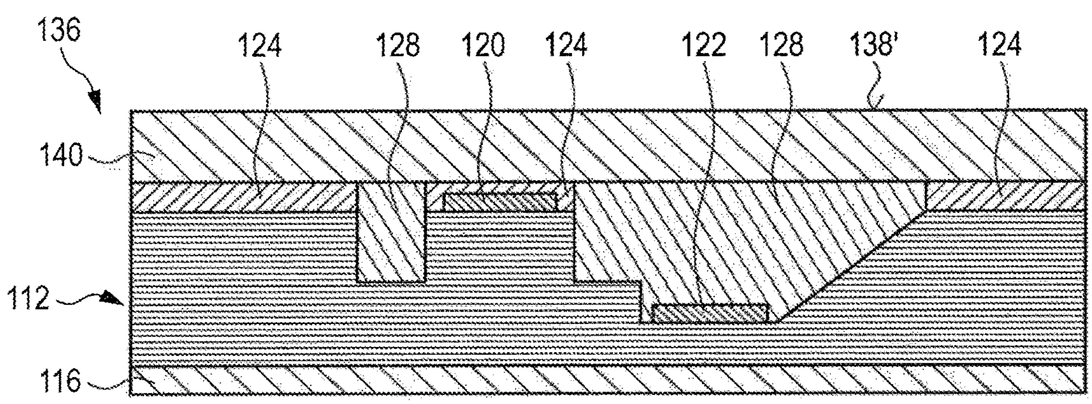


Fig.1D

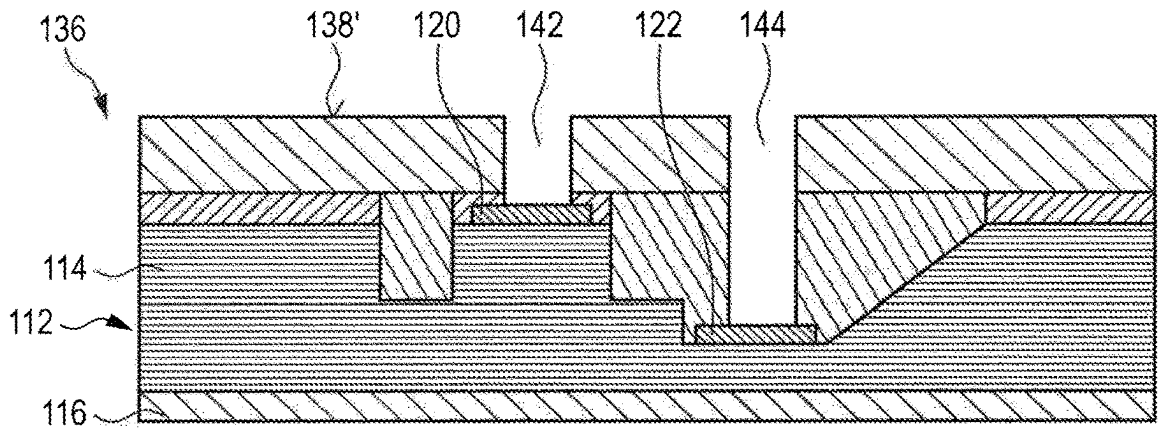


Fig. 1E

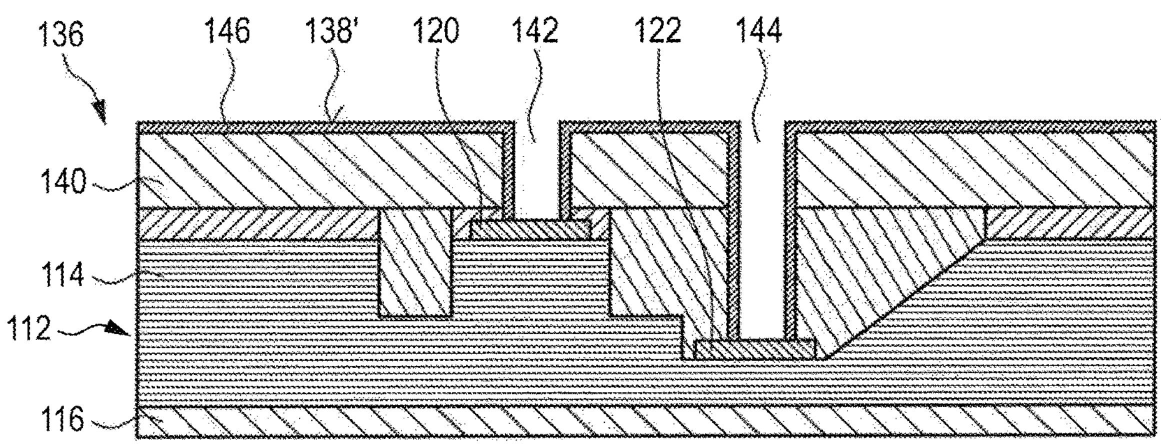


Fig. 1F

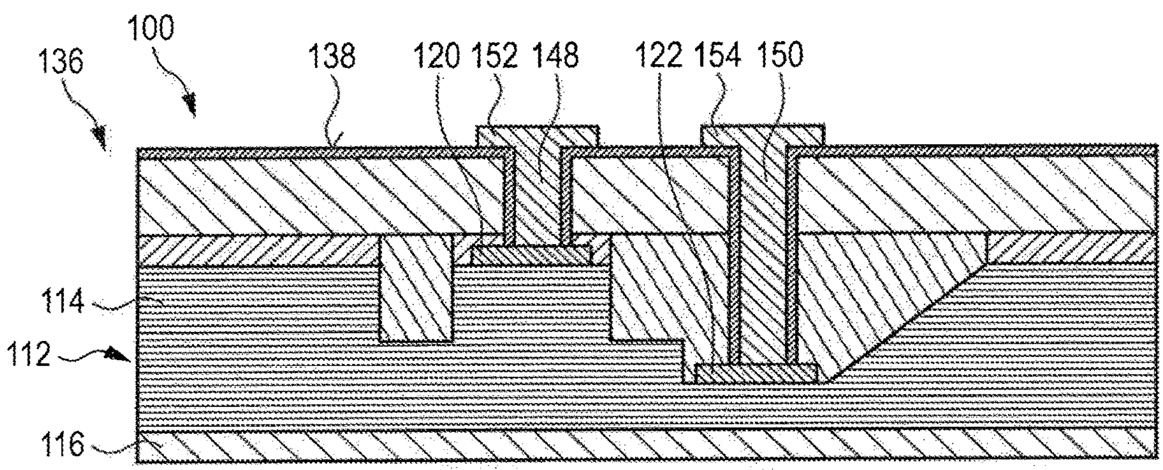


Fig. 1G

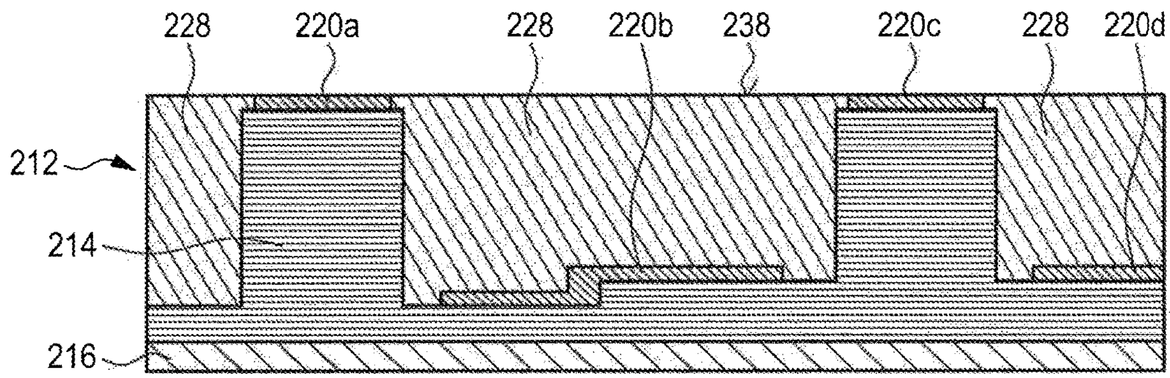


Fig. 2A

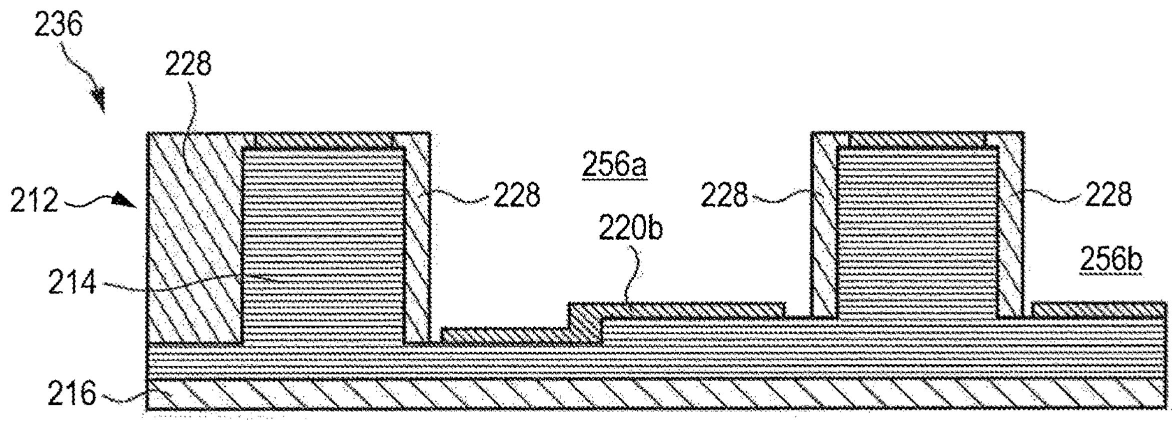


Fig. 2B

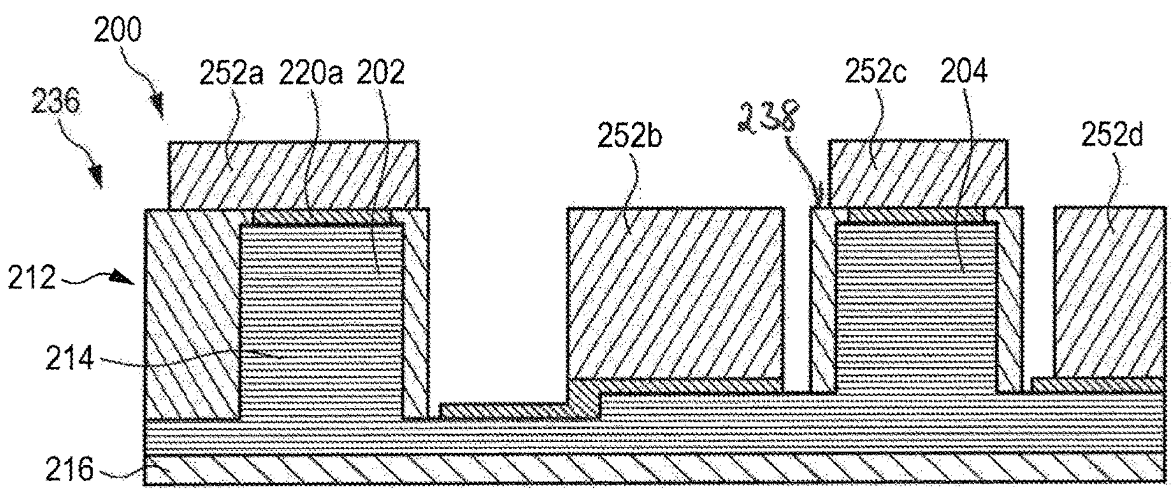


Fig. 2C

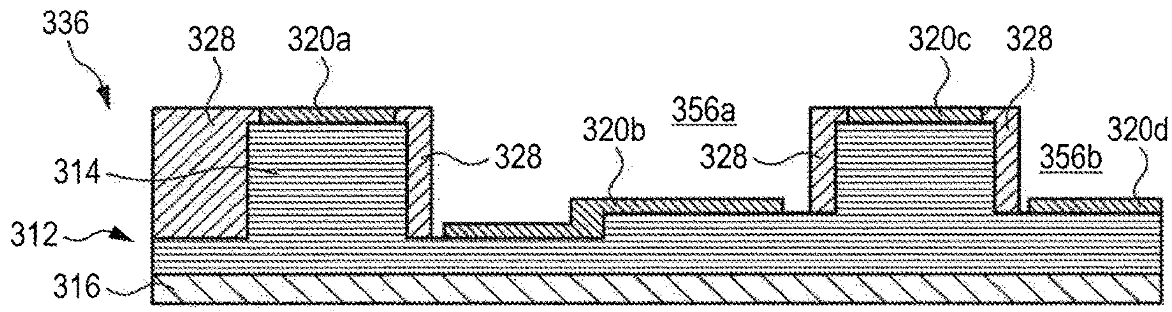
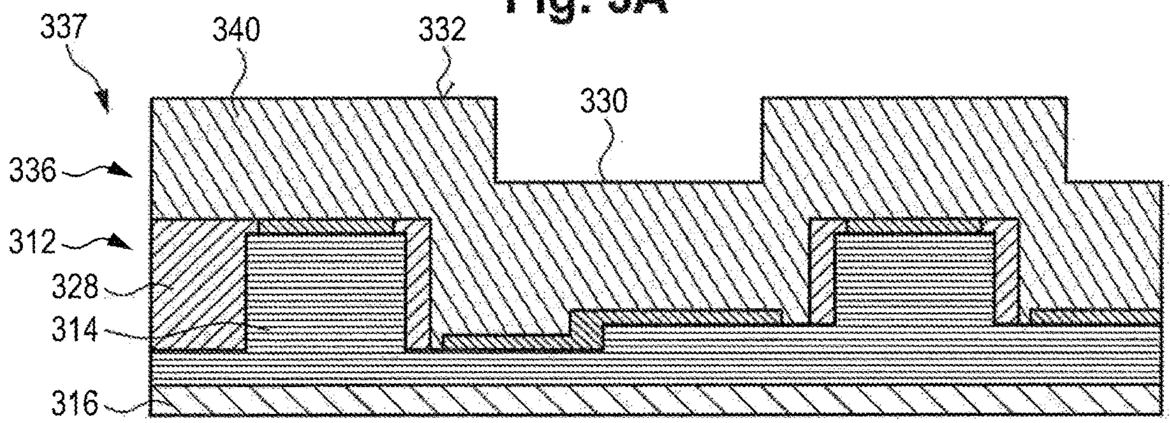


Fig. 3A



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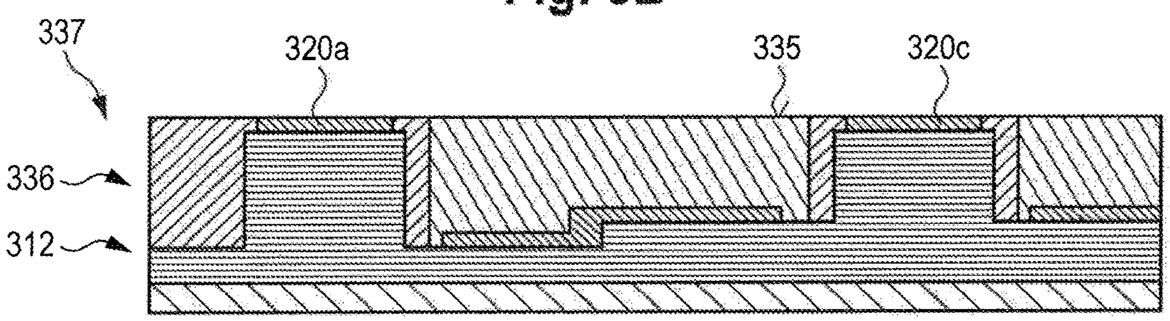

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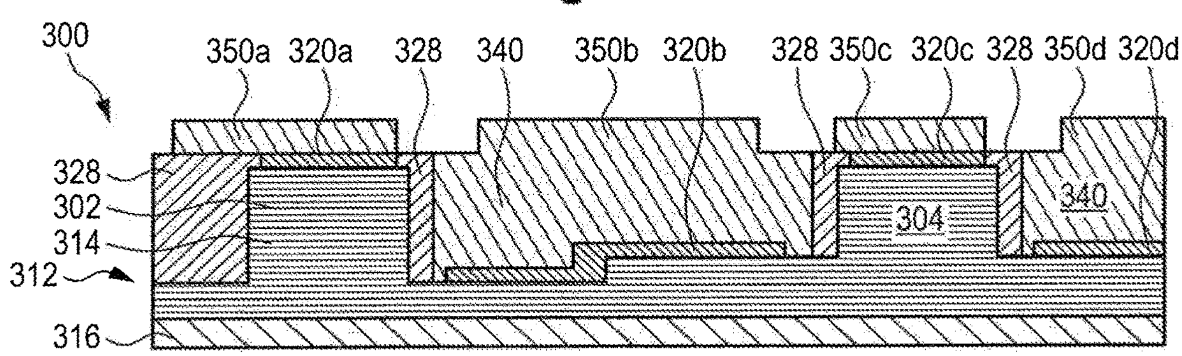

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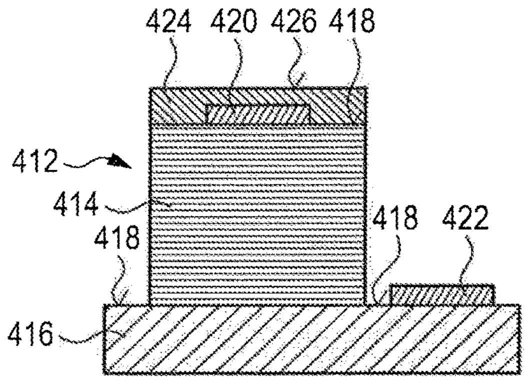


Fig. 4A

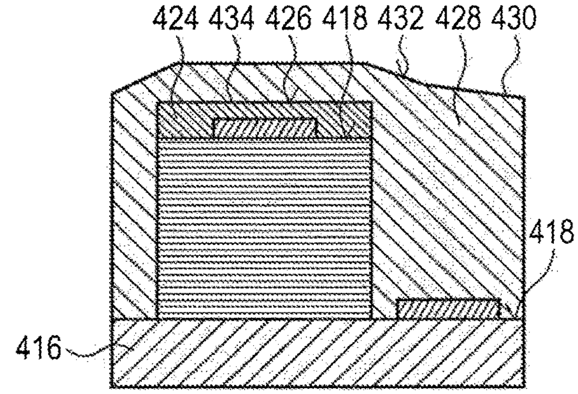
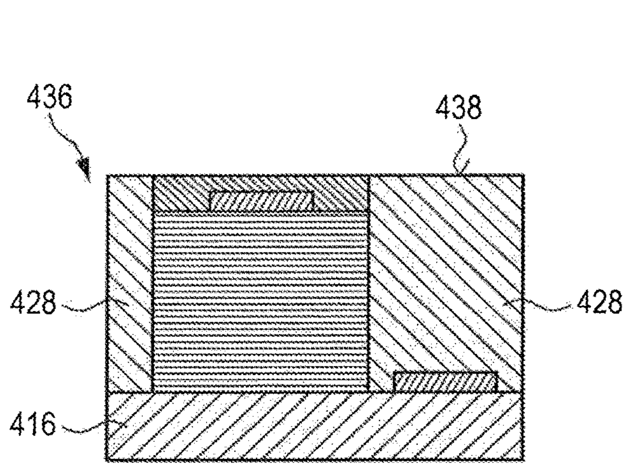


Fig. 4B



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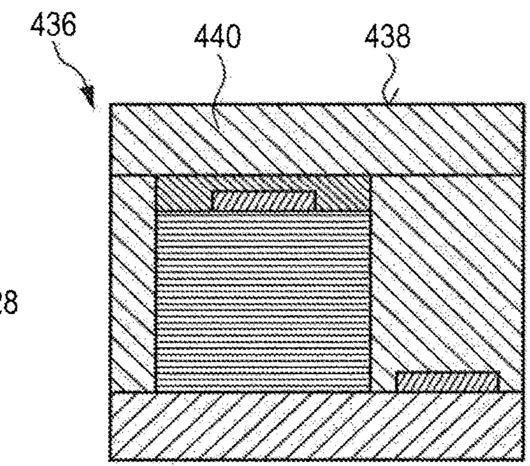

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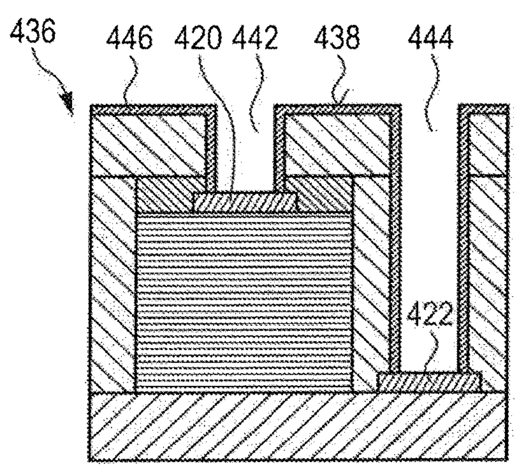
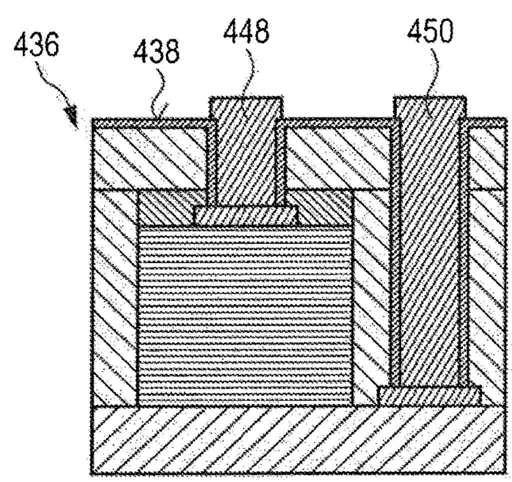


Fig. 4E



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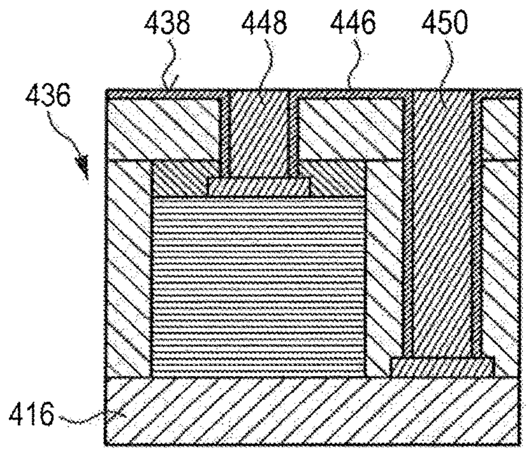

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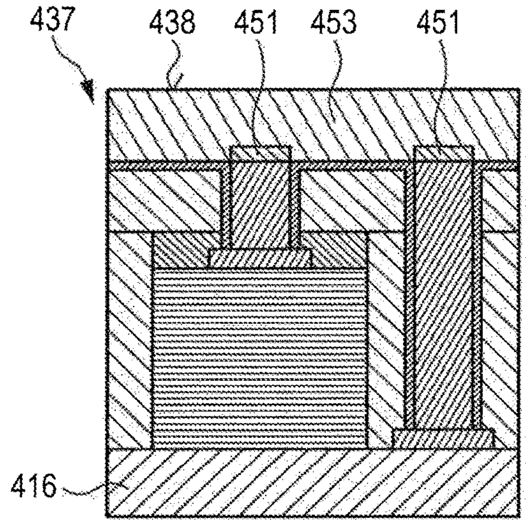

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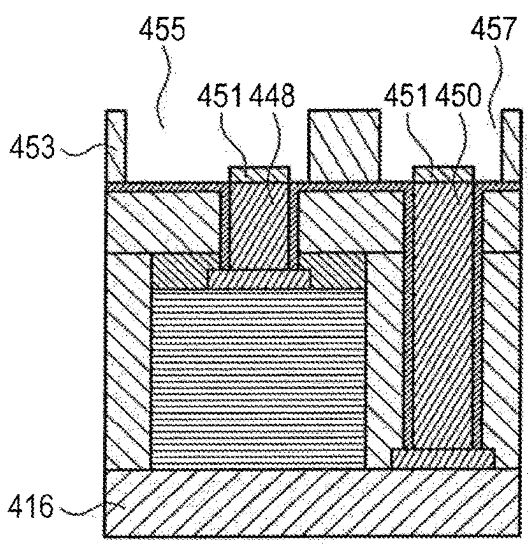

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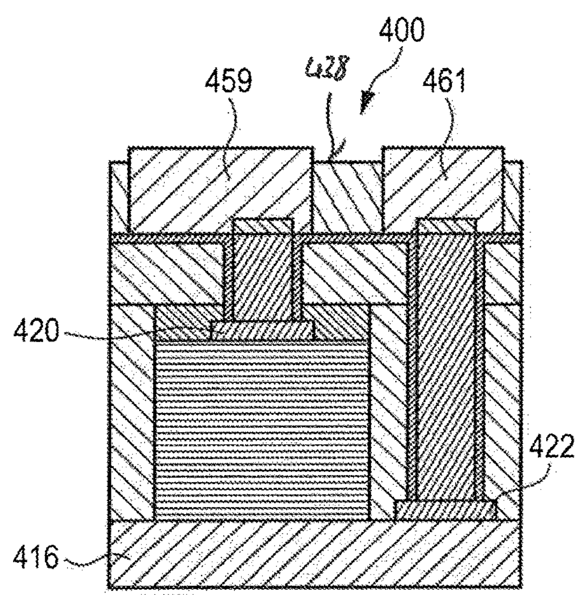

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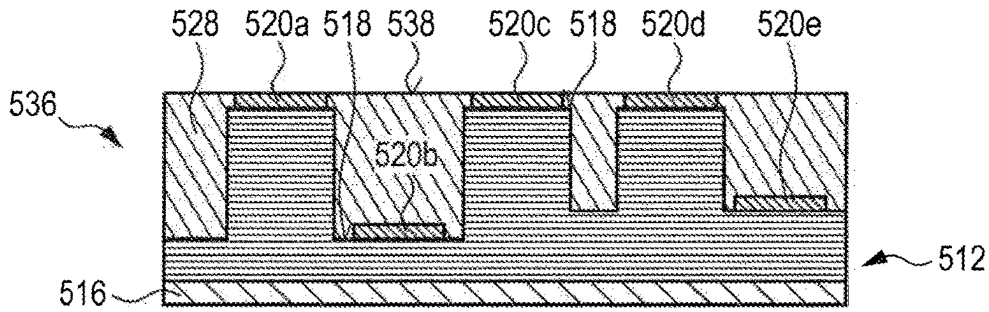
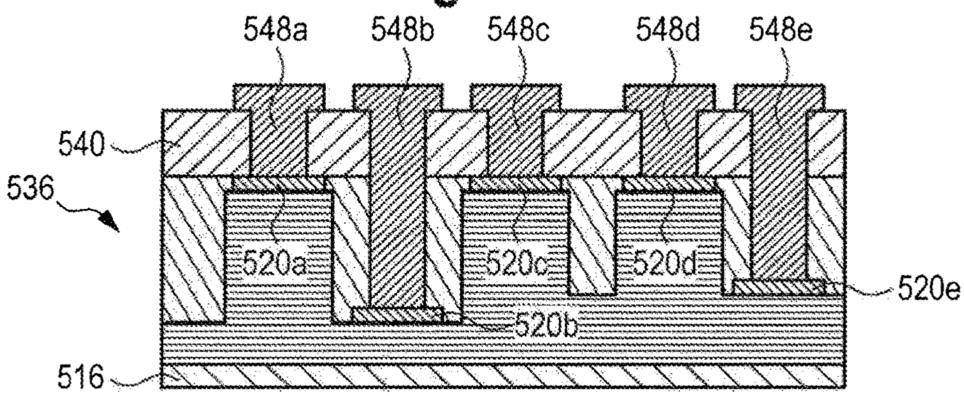


Fig. 5A



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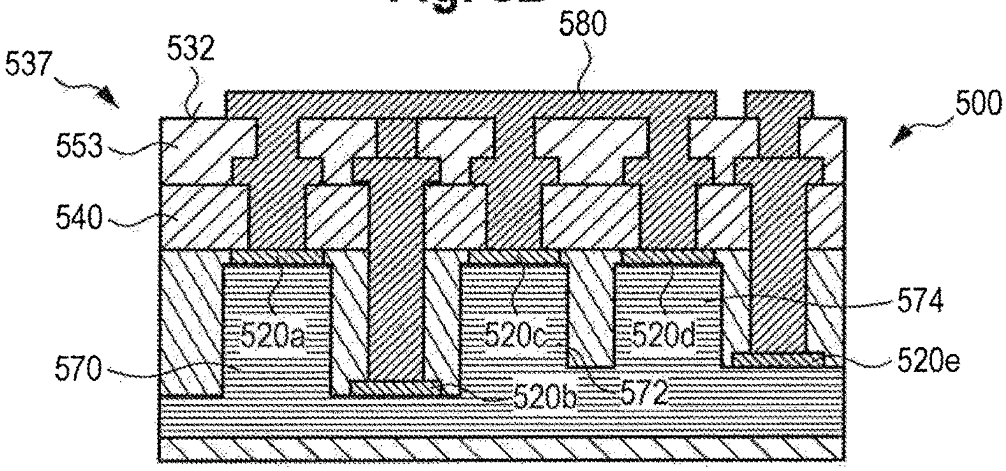

Fig.5C

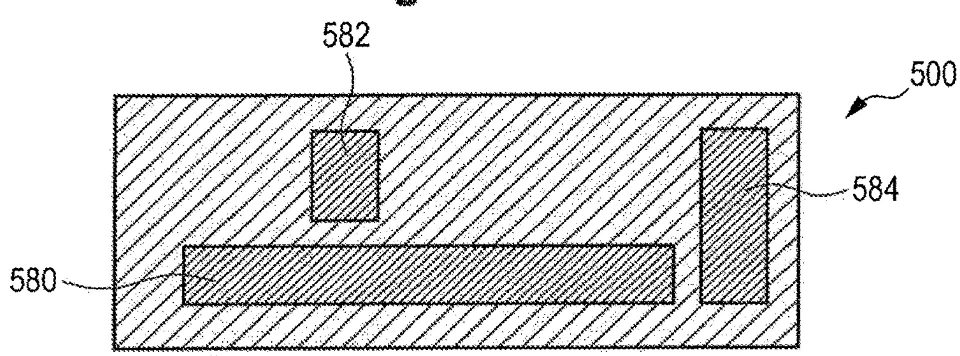

Fig.5D

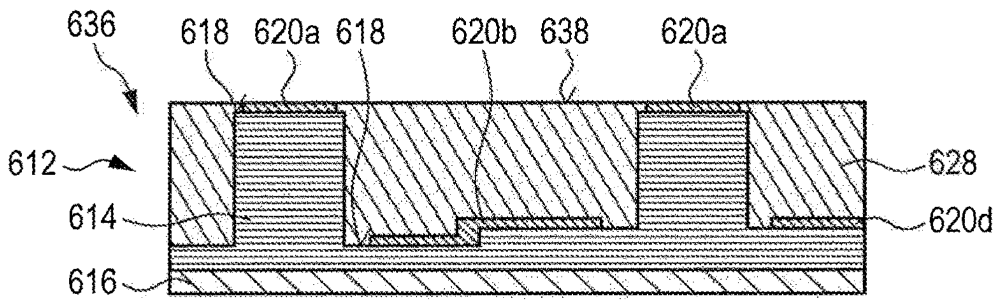

Fig. 6A

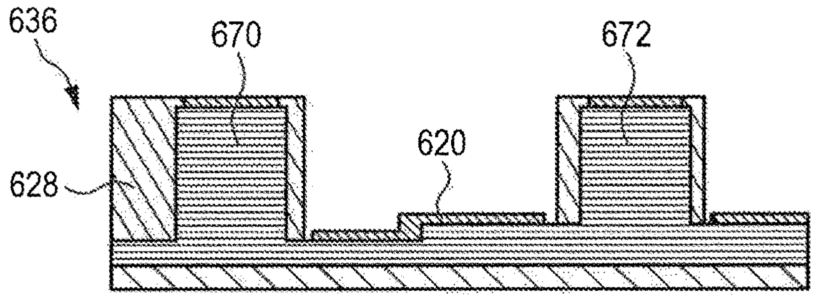

Fig. 6B

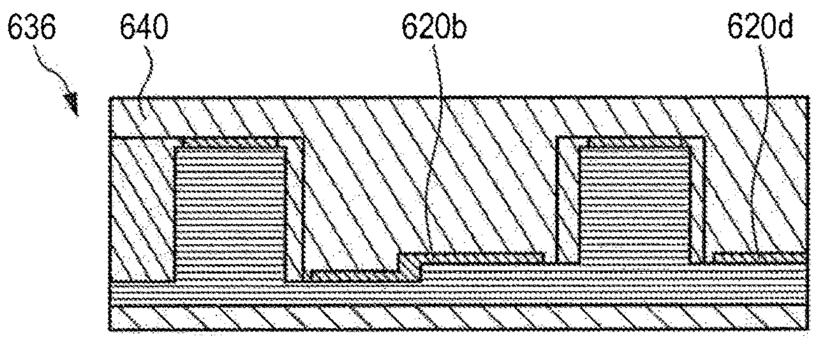

Fig.6C

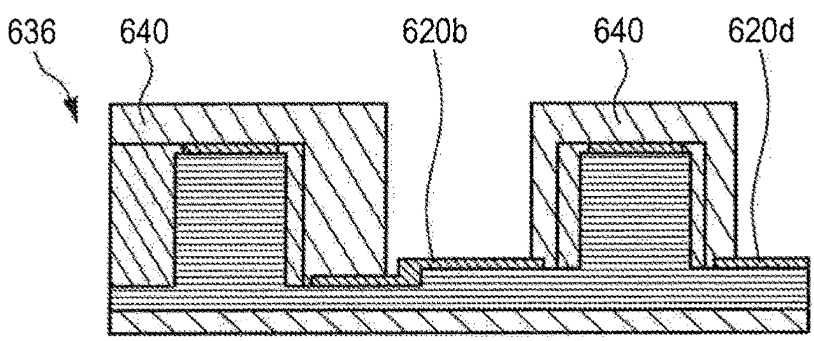

Fig.6D

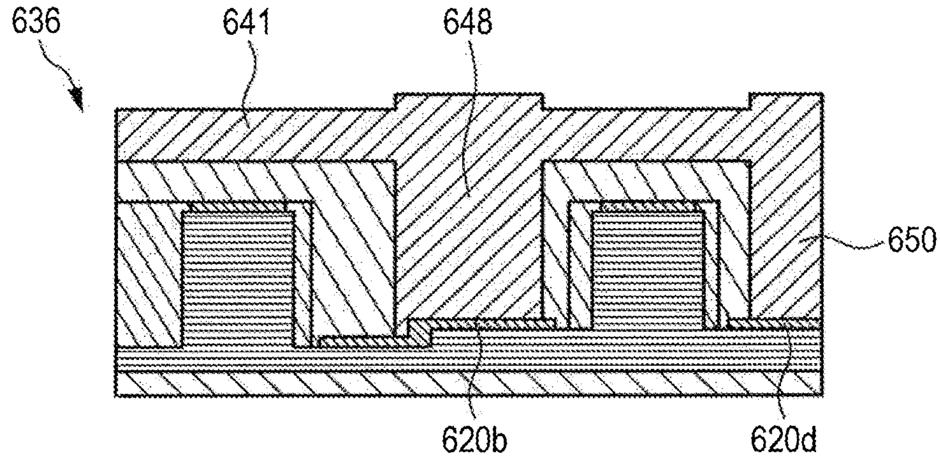

Fig. 6E

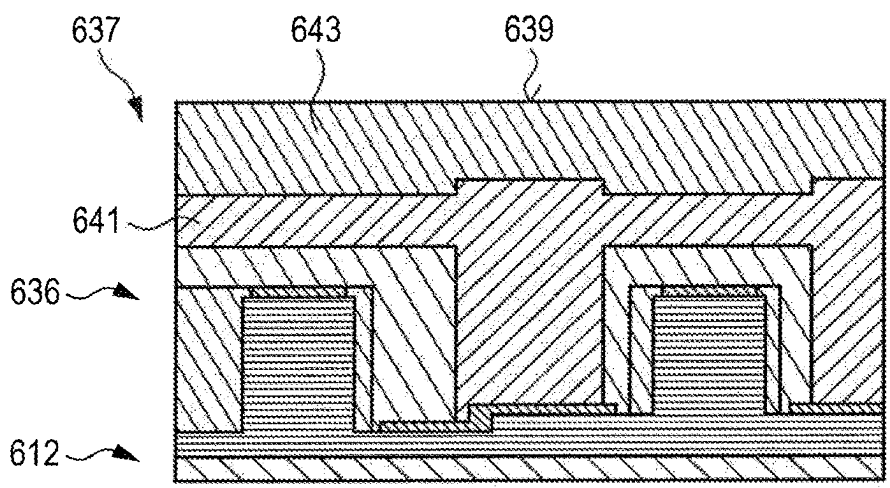

Fig. 6F

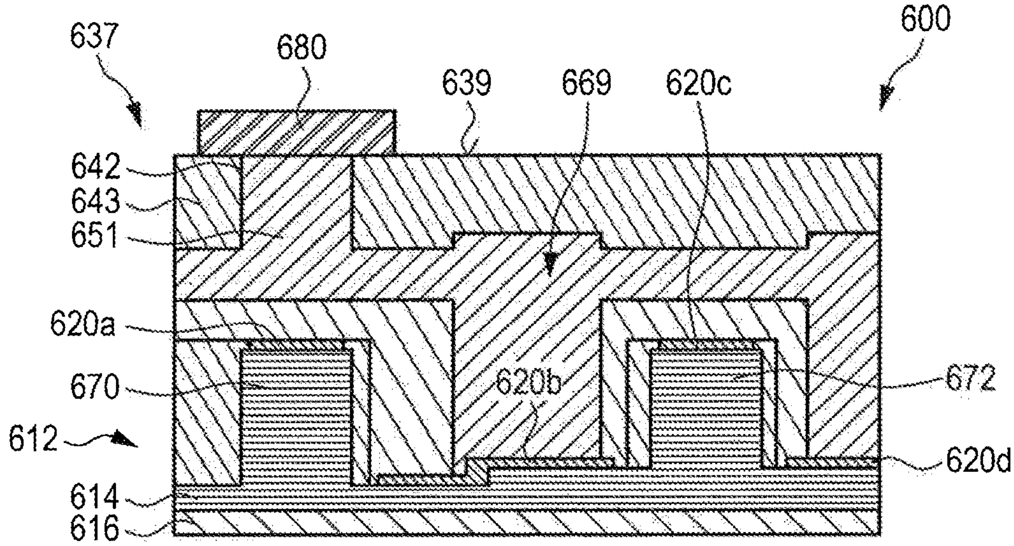
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Fig.6G

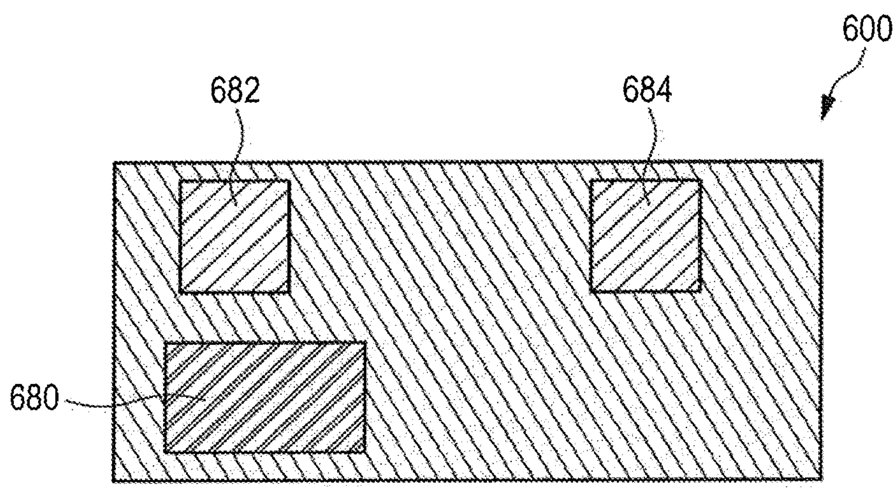
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Fig.6H

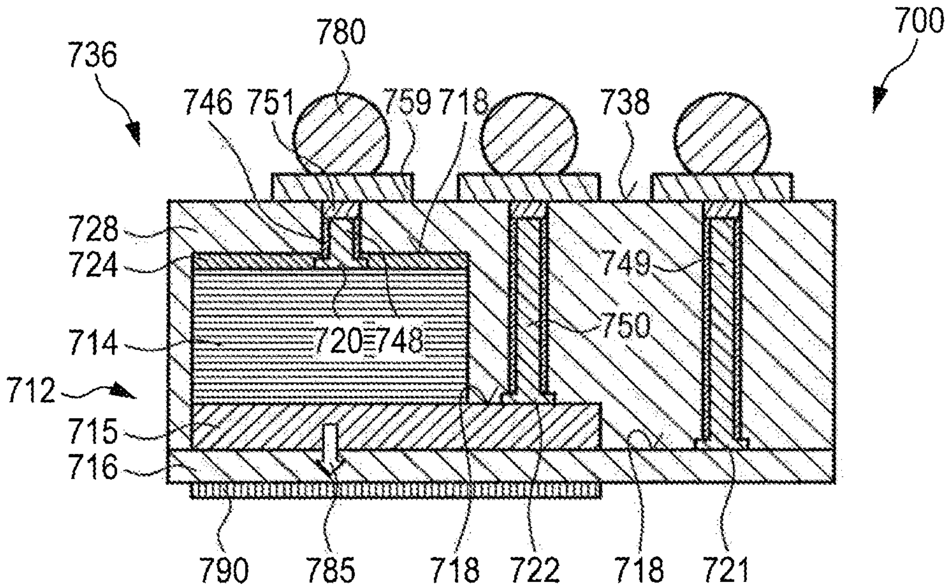
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Fig. 7

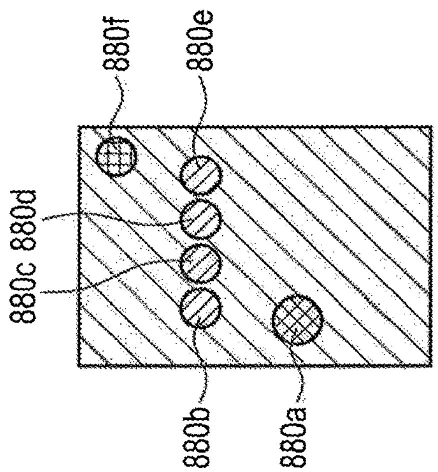
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Fig. 8B

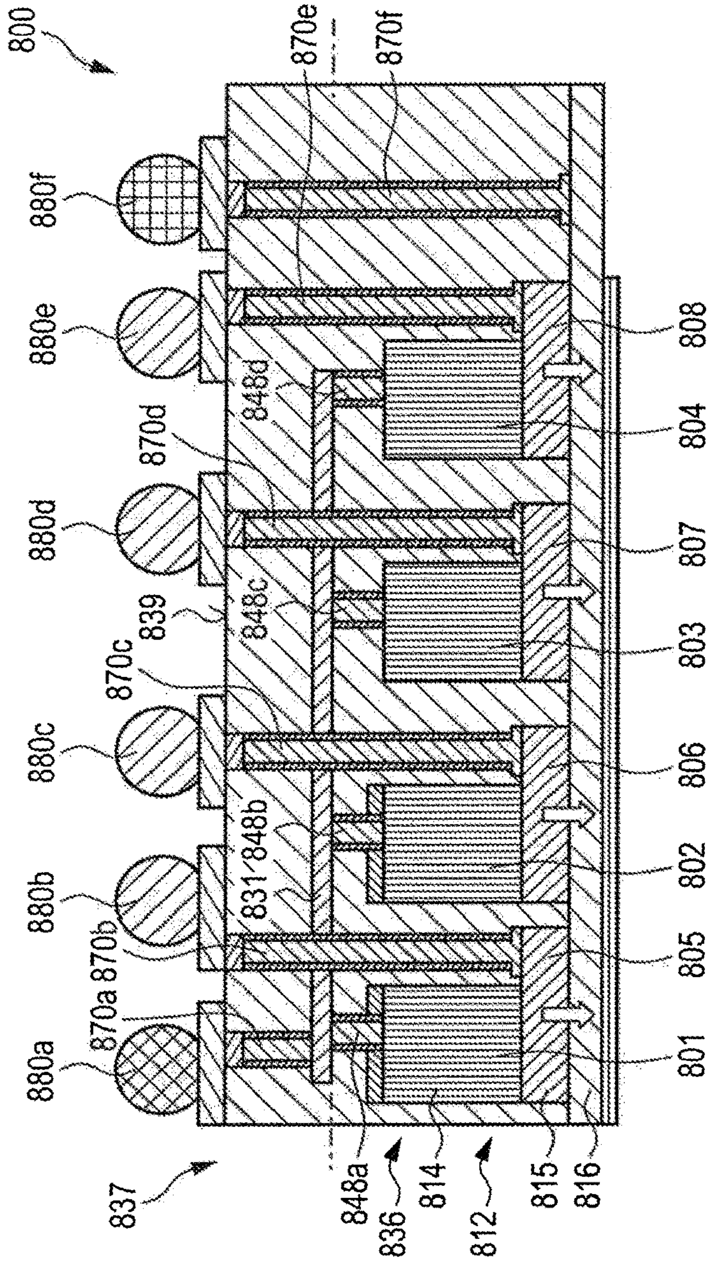
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Fig. 8A

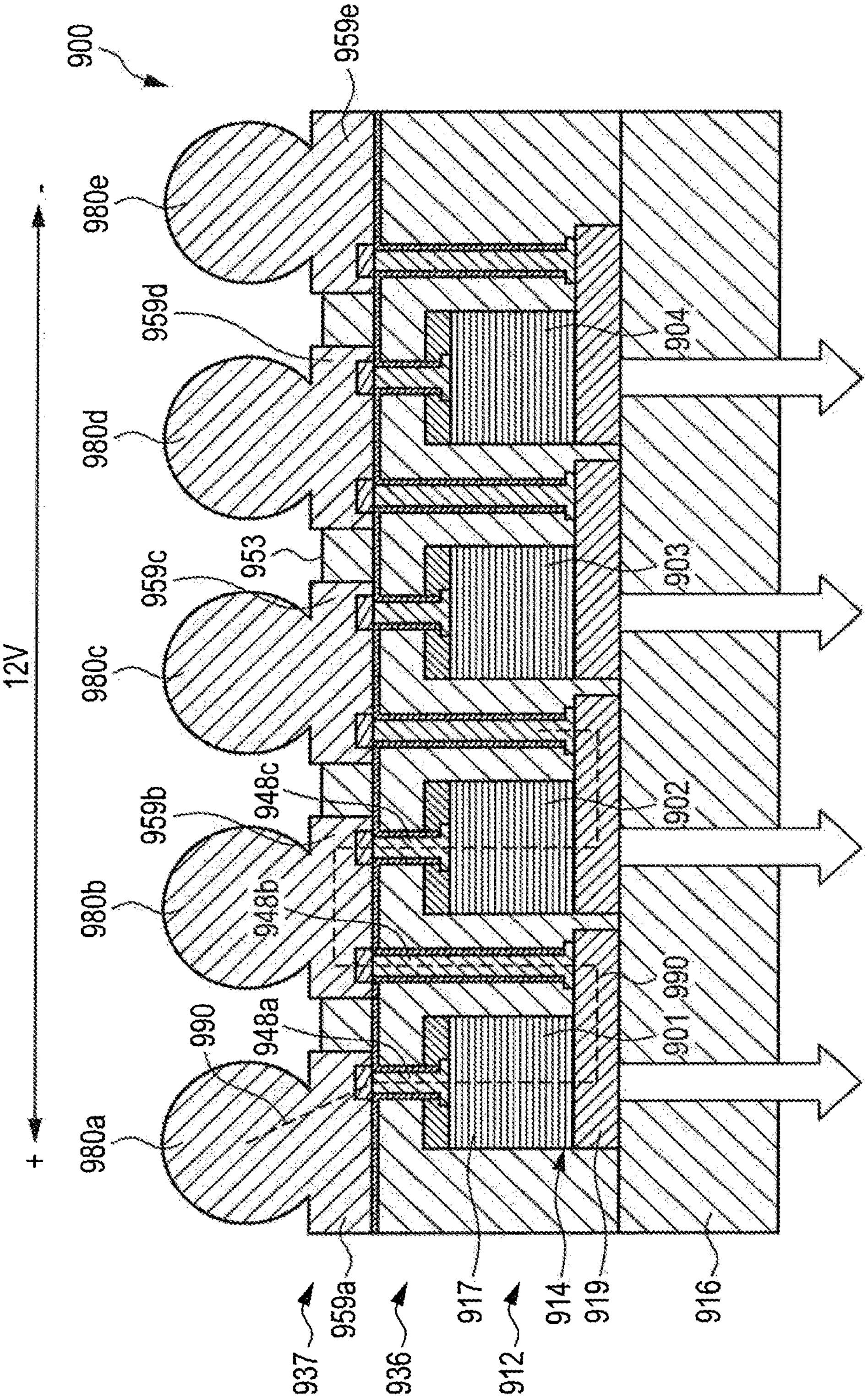
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Fig. 9

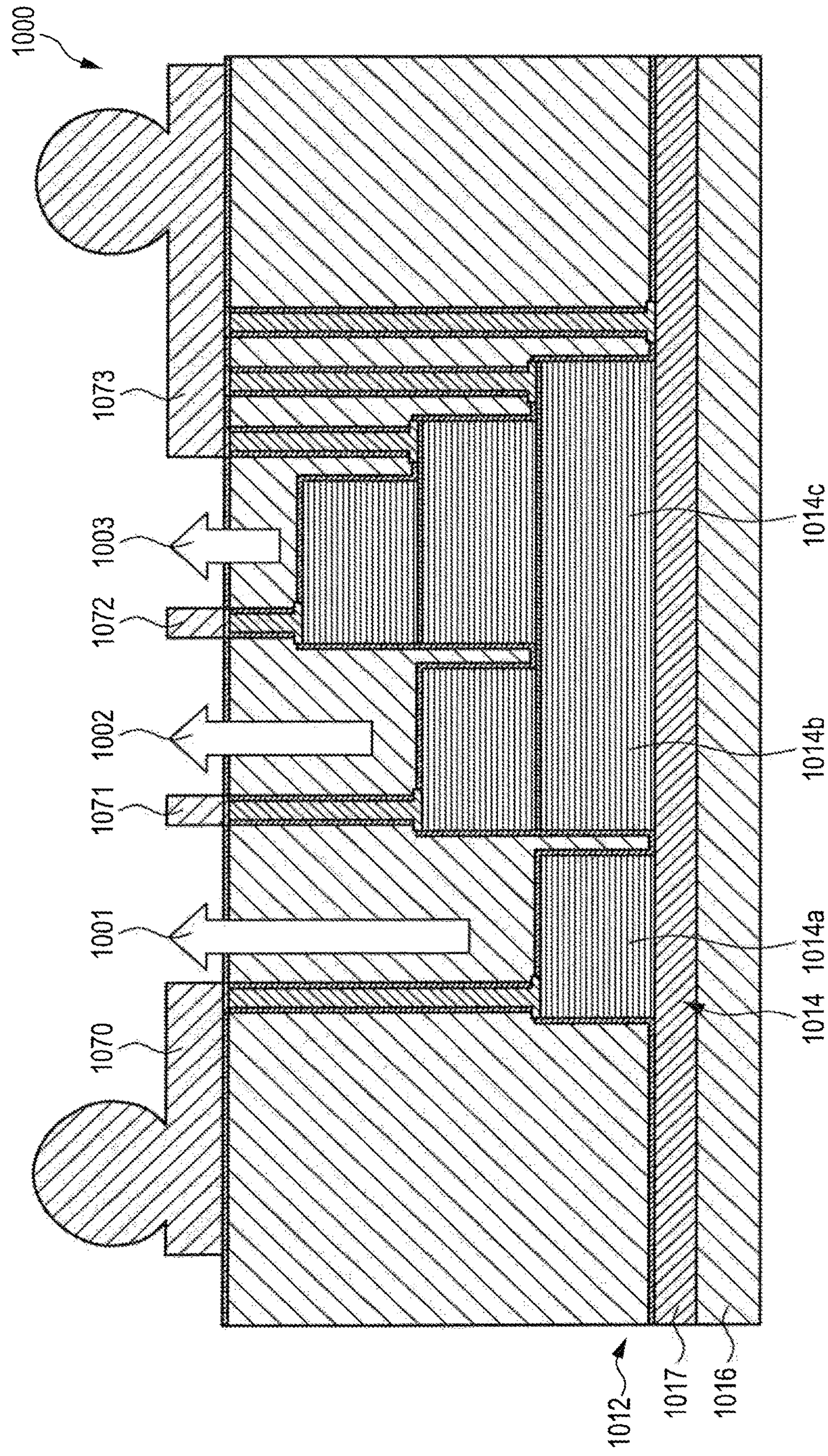

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Fig. 10

METHOD OF FABRICATING A VCSEL DEVICE AND VCSEL DEVICE

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation of International Application No. PCT/EP2020/078400 (WO 2021/074027 A1), filed on Oct. 9, 2020, and claims benefit to European Patent Application No. EP 19203671.3, filed on Oct. 16, 2019. The aforementioned applications are hereby incorporated by reference herein.

FIELD

Embodiments of the present invention relate to a method of fabricating a Vertical Cavity Surface Emitting Laser (VCSEL) device, and to a VCSEL device.

BACKGROUND

VCSELs are a type of semiconductor laser diodes with laser beam emission perpendicular to the top or bottom surface. Typically, a VCSEL comprises two distributed Bragg reflector mirrors parallel to the wafer surface, and an active region comprising one or more quantum wells for the laser light generation arranged between the two distributed Bragg reflector mirrors. The DBR-mirrors typically comprise layers with alternating high and low refractive indices. In common VCSELs, the upper and lower mirrors are doped as p-type and n-type materials, forming a diode junction. In other conventional configurations, the p-type and n-type regions may be embedded between the DBRs. The overall VCSEL layer structure comprises one or more semiconductor materials. When fabricating a VCSEL device comprising one or more VCSELs (VCSEL array), the VCSEL layer structure is epitaxially grown on a wafer. The main part of the VCSEL fabrication process is the electrical isolation of one or more single VCSELs on the wafer. This is typically done by etching the VCSEL epitaxial structure, separating the p-n junction and thus creating a certain topology on the wafer. Depending on the type of VCSEL device, the semiconductor etching can be done multiple times at different points in the process sequence, creating topologies of the top surface of the wafer including the VCSEL layer structure with altitudes up to 15 μm. In order to inject carriers into the VCSEL, electrical contact areas to the n-type doped and p-type doped sides of the p-n junction have to be applied. This is conventionally done by depositing electrically conductive materials at different altitudes on the wafer including the VCSEL layer structure. To provide connection between the VCSEL device and the electrical driver, external electrical links need to be applied. This can be done directly by soldering, while the contact areas need to have a certain size, in order to sufficiently mount the soldering ball on the contact areas. However, VCSEL sizes tend to be smaller (20-30 μm) than a typical solder ball (50-60 μm). Moreover, the VCSEL itself is mechanical unstable, e.g. due to the previous oxidation process for forming a current aperture, thus the direct soldering on the VCSEL structure is impossible. For this reason, it has been necessary to reserve a certain bonding area. In the bonding area, the mechanical stability for the bonding process is given. The bonding area and the electrical contact areas are connected by electrical links on the VCSEL. While this layout enables different bonding techniques like soldering, bumps, etc., a disadvantage is that the VCSEL device fabricated in this way is

multiple times larger in size than the actual VCSEL or VCSELs. This is because the metal tracks need to overcome the surface topology of the device, connecting the contact areas at multiple altitudes. Advanced routing concepts, e.g. to connect only certain VCSELs in a VCSEL array, require a large amount of space, additional electrical isolation and metal layers. The above-mentioned problems have been solved in common fabrication methods by reserving space on the VCSEL chip for soldering purposes, referred to as the bonding area. Electrical connection from the bonding areas to different altitudes of the VCSEL device are done by metal tracks. These tracks have to overcome the altitudes on the wafer including the VCSEL layer structure by graded etching profiles or specialized deposition processes. Thus, a state of the art VCSEL device requires ¼ of the footprint for the VCSEL itself, but ¾ of the footprint for bonding areas, metal tracks and support structures.

Thus, there is a need in an improved fabrication method and an improved VCSEL device.

SUMMARY

Embodiments of the present invention provide a method of fabricating a Vertical Cavity Surface Emitting Laser (VCSEL) device. The method includes providing a first structure comprising a VCSEL layer structure on a wafer. The first structure including the wafer comprising one or more semiconductor materials. The first structure has a non-planar first structure top surface with varying height levels along the non-planar top surface. The non-planar first structure top surface includes one or more electrical contact areas at different height levels above the wafer. The method further includes applying one or more layers of cover material different from the one or more semiconductor materials on the non-planar first structure top surface with a thickness such that a lowest height level of a cover material top surface is equal to or above the highest height level of the non-planar first structure top surface, to obtain a second structure comprising the first structure and the one or more layers of cover material. The second structure has a second structure top surface. The method further includes planarizing the second structure top surface, and producing one or more first electrical vias from the second structure top surface through the one or more layers of cover material for electrical connection to the one or more electrical contact areas.

BRIEF DESCRIPTION OF THE DRAWINGS

Subject matter of the present disclosure will be described in even greater detail below based on the exemplary figures. All features described and/or illustrated herein can be used alone or combined in different combinations. The features and advantages of various embodiments will become apparent by reading the following detailed description with reference to the attached drawings, which illustrate the following:

FIGS. **1A-1G** schematically show side views of a process sequence of an embodiment of a method of fabricating a VCSEL device, wherein FIG. **1G** shows the fabricated VCSEL device in a side view;

FIGS. **2A-2C** schematically show side views of a processing sequence of another embodiment of a method of fabricating a VCSEL device, wherein FIG. **2C** shows the fabricated VCSEL device in a side view;

FIGS. **3A-3D** schematically show side views of a processing sequence of another embodiment of a method of fabricating a VCSEL device, wherein FIG. **3**D shows the fabricated VCSEL device in a side view;

FIGS. **4**A-**4**J schematically show side views of a processing sequence of another embodiment of a method of fabricating a VCSEL device, wherein FIG. **4**J shows the fabricated VCSEL device in an side view;

FIGS. **5**A-**5**D schematically show a processing sequence of another embodiment of a method of fabricating a VCSEL device, wherein FIG. **5**A-**5**C show side views and FIG. **5**D shows the fabricated VCSEL device in a top view;

FIGS. **6**A-**6**H schematically show a processing sequence of another embodiment of a method of fabricating a VCSEL device, wherein FIG. **6**A-**6**G show side views and FIG. **6**H shows the fabricated VCSEL device in a top view;

FIG. **7** show a side view of another embodiment of a VCSEL device fabricated according to embodiments of the present invention;

FIGS. **8**A-**8**B show a VCSEL device fabricated according to embodiments of the present invention; wherein FIG. **8**A is a side view and FIG. **8**B is a top view;

FIG. **9** shows a side view of a VCSEL device fabricated according to embodiments of the present invention; and

FIG. **10** shows a side view of a VCSEL device fabricated according to embodiments of the present invention.

DETAILED DESCRIPTION

Embodiments of the present invention provide a fabrication method which enables fabrication of a VCSEL device having a reduced footprint.

According to a first aspect, a method of fabricating a vertical cavity surface emitting laser (VCSEL) device includes:

- providing a first structure comprising a VCSEL layer structure on a wafer, the first structure including the wafer comprising one or more semiconductor materials, the first structure having a non-planar first structure top surface with varying height levels along the non-planar top surface, wherein the non-planar first structure top surface comprises one or more electrical contact areas at different height levels above the wafer;
- applying one or more layers of cover material different from the one or more semiconductor materials on the non-planar first structure top surface with a thickness such that a lowest height level of a cover material top surface is at least equal to or above the highest height level of the non-planar first structure top surface, to obtain a second structure comprising the first structure and the one or more layers of cover material, the second structure having a second structure top surface;
- planarizing the second structure top surface;
- producing one or more first electrical vias from the second structure top surface through the one or more layers of cover material for electrical connection to the one or more electrical contact areas.

The method may be carried out in a different order than indicated above. The method may comprise further processing steps before, between and after the steps indicated above. The method according to the invention proposes a new concept of fabricating VCSEL devices which results in VCSEL devices with smaller footprint. The method according to the invention provides to planarize the non-planar top surface of the initial structure of wafer and VCSEL layer structure. Planarization is performed by applying one or more layers of cover material onto the non-planar top surface of the initial (first) structure. The cover material is then worked to provide a planarized second structure comprising the initial VCSEL wafer structure and the one or more layers of cover material. Thus, the second structure has a planar top surface. One or more electrical bonding areas for externally connecting the VCSEL device with a driver, may be arranged directly on top of the planarized VCSEL chip in electrical contact with the electrical vias, thereby reducing the footprint significantly.

The method according to embodiments of the present invention is not only applicable for VCSEL devices with a single mesa or VCSEL, but also for VCSEL arrays having multiple mesas. Embodiments of the present invention enable single-addressable VCSELs in an array on one chip.

Further, the planarization allows for improving heat-transfer by direct-external connection to a heat sink. Furthermore, a VCSEL device fabricated according to embodiments of the present invention allows for shorter electrical connection between the VCSEL device and an external driver. A shorter electrical connection reduces parasitic capacitances. A further advantage is that internal routing of electrical links within the VCSEL device. i.e. below the top surface of the VCSEL device, is enabled by repeating the planarization sequence multiple times, as will be described herein.

A further advantage is the improved mechanical stability of the VCSEL device due to the cover material.

The method according to embodiments of the present invention may start with an electrically functional VCSEL device on wafer. The VCSEL layer structure comprises one or more semiconductor materials, preferentially II-VI or III-V compound semiconductor materials. For example, the wafer may be a GaAs wafer, and the VCSEL layer structure may comprise GaAs and AlGaAs or InGaAs layers. The cover material or cover materials may be chosen such that it is suitable for being worked, preferentially for being polished, in particular chemically mechanically polished using a slurry. The slurry may contain small particles of the cover material(s).

The VCSEL layer structure may be epitaxially grown on the wafer and then be etched to produce the first structure. The VCSEL layer structure may comprise distributed Bragg reflectors, one or more active regions comprising one or more quantum wells, one or more integrated photodiode layer or phototransistor layer structures. The one or more electrical contact areas are provided for electrically contacting layers of the VCSEL layer structure with different polarities, e.g. for contacting p-regions and n-regions of the VCSEL layer structure.

The second structure top surface may be the outermost surface of the final VCSEL device, or may be an intermediate surface, when the application of cover material layers and planarization is repeated one or more times.

In the following, various embodiments of the method according to invention will be described.

In an embodiment, the planarizing comprises polishing, in particular chemical mechanical polishing the second structure top surface.

The method may further comprise, prior to applying the one or more layers of cover material, applying a working stop layer. Such a working stop layer may be applied on the non-planar first structure top surface preferentially in areas with the highest height levels of the VCSEL layer structure. The working stop layer advantageously avoids unwanted removal of material from the VCSEL layer structure.

The working stop layer preferentially comprises a material which is different from the cover material to be worked so that, when working, e.g. polishing the cover material with a slurry containing small particles of the cover material, the slurry does not remove material from the working (polishing) stop layer.

During working the cover material, the wafer may be fixed from the back side by an adhesive tape, vacuum mounting or by arranging it on an appropriate carrier substrate to prevent mechanical damage to the wafer.

In an embodiment, the planarization may comprise applying a first layer of cover material, working the first cover material top surface to provide a planar first cover material top surface, and applying a second layer of cover material on the planar first cover material top surface to provide a second cover material top surface.

The second cover material may be a electrically isolating and thermally conducting. The second cover material layer may provide electrical isolation of the VCSEL chip or wafer and, in particular, mechanical stability. The second layer preferentially is sufficiently thick, e.g. 100-200 nm, to provide sufficient mechanical stability, but should not be too thick in order to avoid a high altitude of the VCSEL device.

In some embodiments, at least one of the layers of cover material may be electrically isolating. The advantage here is that the cover material may also provide an electrical isolation of areas or regions of the VCSEL device having different polarities, and/or provides electrical isolation between mesas of a VCSEL array device.

In other embodiments, at least one of the layers of cover material may be metallic, and thus, electrically conductive. By working, e.g. polishing the metal cover material with a suitable polishing agent, planarization may be performed as it is the case with an electrically isolating cover material. An advantage of using a metal as the cover material is that at least part of vias from the final top surface of the VCSEL chip to the contact areas lying on lower height levels may be provided by the metal layer so that processing may be simplified.

It is to be understood, that applying one or more layers of metallic cover material on the non-planar first structure top surface may be combined with application of one or more layers of electrically-isolating cover material.

Further, at least one of the layers of cover material may be thermally conducting. Thus, heat discharge and connection to a heat sink may be simplified and more effective.

The first electrical vias may be produced by etching one or more contact holes into the one or more layers of cover material down to the one or more electrical contact areas and filling the one or more contact holes with an electrically conducting material, e.g a metal, up to the second structure top surface.

Producing the contact holes may be carried out by etching the second structure, e.g. by plasma assisted dry etching (RIE/ICP), thereby exposing the one or more electrical contact areas on the bottom of the contact holes. The etching chemicals are thereby not attacking the bottom contact areas, thus creating a self-terminating edging process. A cleaning step may follow for cleaning the contact hole openings, using e.g. wet-chemical cleaning with HCL or H_2SO_4, or plasma cleaning with $O_2/Ar/NH_3$.

Filling the contact holes with an electrically conducting material may be performed galvanically. To do so, a metal film serving as a galvanic seed layer may be applied to provide electrical conductivity for the galvanic contact hole filling. Before galvanically filling the contact holes, the contact holes may be filled up with a protective coating, e.g. by atomic layer deposition or sputtering to create a layer on the walls of the contact holes with a thickness of several nm. This layer may be advantageous as it may absorb residual stress created by the galvanic contact hole filling. Further, it may provide adhesion and inhibit diffusion of material from optional subsequent application of further material(s) onto the second structure.

The method according to embodiments of the present invention has the further advantage of allowing forming electrical links or electrical routings within and on top of the VCSEL chip. In particular, complex internal routing of electrical links can be performed by repeating the planarization sequence two or more times. This allows e.g. for photonic component (like photodiode or phototransistor) integration in the VCSEL device, for electrical connection of a part or all of a plurality of mesas of the VCSEL device with one another, and/or to provide multi-level connections for tunnel diodes, coupled active regions, intra-cavity components and the like.

In an embodiment, the method according to embodiments of the present invention may further comprise, after producing the one or more first electrical vias, applying at least one further layer of cover material on the second structure to provide a third structure comprising the second structure and the further cover material, the third structure having a third structure top surface. In this embodiment, the second structure top surface is an intermediate surface of the VCSEL device.

The method may further comprise producing further electrical vias through the planar third structure top surface down to at least a part of the first electrical vias for electrically connecting the one or more further electrical vias.

In the context of the afore-mentioned embodiments, the method may further comprise, prior to applying the further cover material, electrically connecting at least a part of the first electrical contacts on the planar first structure top surface with one another. Thereby, an internal electrical routing between e.g. multiple mesas arranged on the VCSEL chip is enabled.

The afore-mentioned processing steps may be repeated several times, in order to create internal electrical connections, e.g. between multiple mesas, or photonic components integrated into the VCSEL chip in a multitude of height levels above the wafer. Thus, the afore-mentioned embodiments are particularly advantageous for producing VCSEL arrays having a multitude of VCSELs or mesas. Embodiments of the present invention allows to make internal connections in the VCSEL array, e.g. to connect mesas with one another, while the external link to the driver is executed directly above the VCSEL chip, thus creating dense packet VCSEL arrays with only a low number of contacts or bond areas being necessary.

According to a second aspect, a Vertical Cavity Surface Emitting Laser (VCSEL) device includes:

- a first structure comprising a VCSEL layer structure on a wafer, the first structure including the wafer comprising one or more semiconductor materials, the first structure having a non-planar first structure top surface with varying height levels along the non-planar top surface, wherein the non-planar top surface comprises one or more electrical contact areas at different height levels;
- one or more layers of cover material different from the one or more semiconductor materials arranged on the non-planar first structure top surface, wherein an uppermost top surface of the VCSEL device is planar,
- one or more first electrical vias from the uppermost top surface through the one or more layers of cover material in electrical connection with the one or more contact areas.

The VCSEL device according to embodiments of the present invention has the same or similar advantages as the method according to the first aspect.

In particular, the VCSEL device according to embodiments of the present invention has a footprint which is significantly reduced in comparison with conventional VCSEL devices, e.g. by a factor of 2 or more. The VCSEL device according to embodiments of the present invention may comprise a single mesa, or a plurality of mesas to provide a VCSEL array which is densely packed. Other configurations like integration of photodiode/phototransistor structures are also possible, as well as VCSEL devices with multi-wavelength emission.

The one or more layers of cover material may include one or more electrically isolating layers and/or one or more electrically conductive layers.

The VCSEL device may comprise one or more internal electrical links electrically connecting one or more of the vias at a height level below the uppermost top surface and above the wafer.

The VCSEL device may comprise one or more bonding areas on the uppermost top surface in electrical connection with the one or more electrical vias.

The VCSEL device may be a bottom emitter or a top emitter. The VCSEL device may be fully enclosed by an electrical isolating and thermal conductive material formed by the one or more layers of cover material as indicated above.

Further features and advantages will become apparent from the following description of exemplary embodiments with reference to the drawings.

In the following, several embodiments of VCSEL devices and embodiments of a method of fabricating the VCSEL devices will be described.

A first embodiment of a method of fabricating a VCSEL device **100** will be described with reference to FIGS. **1A**-**1G**. FIG. **1G** shows the VCSEL device **100** fabricated according to the method.

According to FIG. **1A**, a first structure **112** is provided. The first structure **112** comprises a VCSEL layer structure **114** on a wafer **116**. The VCSEL layer structure **114** including the wafer **116** comprise one or more semiconductor materials. As an example, the wafer **116** may be a GaAs-wafer, and the VCSEL layer structure **114** may comprise layers of GaAs and AlGaAs. Other semiconductor materials chosen from the group of II-VI or III-V compound semiconductors are conceivable as materials for the first structure **112**.

The VCSEL layer structure **114** may be epitaxially grown on the wafer **116** according to known techniques. The VCSEL layer structure **114** may comprise distributed Bragg reflectors and one or more active regions comprising one or more quantum wells as known in the art. As shown in FIG. **1A**, the VCSEL layer structure **114** may be provided as an electrical functional VCSEL structure, i.e. it is ready for laser emission if connected with a driver. In the process of fabricating a VCSEL device like the VCSEL device **100**, the VCSEL layer structure **114** is etched into the epitaxial layer structure, in order to separate the p-n junction of the VCSEL layer structure **114**, whereby a certain topology of the first structure is created. Accordingly, the first structure **112** has a non-planar first structure top surface **118** with varying height levels along the non-planar top surface **118**. The first structure **112** is provided with one or more electrical contact areas **120**, **122**. The contact areas **20**, **22** may be provided as contact areas with different polarity. Contact area **20** for example may be a p-contact area, and contact area **22** may be a n-contact area. The one or more electrical contact areas **120**, **122** are provided for electrically contacting regions of the VCSEL layer structure **114** of different polarities, e.g. for contacting p-regions and n-regions of the VCSEL layer structure **114**.

The non-planar topology of the surface **118** of the first structure **112** and, thus, the arrangement of the contact areas **120**, **122** in different height levels above the wafer **116** makes it difficult to apply external electrical links for connection with an electrical driver (not shown). In particular, applying solder balls on the contact areas **120**, **122** in the state of the first structure **112** shown in FIG. **1A** would require much space, as soldering balls typically have a size of 50-60 μm, while VCSEL sizes tend to be smaller, for example 20-30 μm. This means that applying external links to the first structure **112** would require ¾ of the VCSEL device size as the footprint for bonding areas, while the VCSEL itself would only make ¼ of the footprint.

The fabrication method described hereinafter solves this problem.

After the final topology of the first structure is reached in the front-end process sequence, the first structure **112** may be covered with a layer of cover material **124**, which preferentially is mechanical stable and electrically isolating. The material **124** is applied along a part of the non-planar first structure top surface **118** as shown in FIG. **1A**. As can be seen in FIG. **1A**, the material **124** is applied on the non-planar first structure top surface **118** in areas where the top surface **118** has the highest height levels. A top surface **126** of the material **124** defines the highest level of the non-planar top surface **118**. The material **124** serves as a stop layer when the top surface **118** is worked, in particular polished, in a subsequent process step which will be described below. The material **24** may be nitride-based, for example AlN or SiN.

Next, according to FIG. **1B**, a layer of cover material **128** different from the semiconductor material or materials of the VCSEL layer structure **114** and wafer **116** is applied on the non-planar first structure top surface **118** including the top surface **126** of the material **124** along the non-planar first structure top surface **118**, **126**. The cover material **128** is applied with a thickness such that a lowest height level **130** of a cover material top surface **132** is at least equal to or above the highest level **134** of the non-planar first structure top surface **118**, **126**, as shown in FIG. **1B**.

In the present embodiment, the cover material **128** is electrically isolating. The cover material **128** may be an oxide-based material, for example Al_2O_3 or SiO_2. The cover material **128** may be applied by conformal sputtering or chemical vapor deposition (CVD), implementing minimal stress into the underlying VCSEL layer structure **114**.

The next step is a planarization step shown in FIG. **1C**. Planarization is performed by working the cover material top surface **132**, whereby a second structure **136** is provided which comprises the first structure **112** and the cover material **128**. Working the cover material top surface **132** may be performed by chemical mechanical polishing using a slurry. The slurry may contain small particles of the cover material **128**. As mentioned above, the material **124** acts as a polishing stop layer so that unwanted removal of material of the VCSEL layer structure **114** is avoided. The second structure **136** comprises a second structure top surface **138** which is planar. In other words, the second structure **136** has a planarized topology in contrast to the non-planar topology of the first structure **112**.

In the present embodiment, a further layer of cover material **140** is applied onto the planar second structure top surface **138**. The cover material **140** may be an electrically isolating material. The thickness of the cover material **140** should be sufficient, e.g. 100-200 nm, to provide mechanical stability, but it should not be too thick to create a high topology above the wafer **116**. The cover material **140** may be nitride-based, and may comprise AlN or SiN, for example. After the cover material **140** is applied, the planar second structure top surface **138** is now formed by the top surface of the cover material **140**, which is labeled with reference numeral **138** again, as indicated in FIG. **1**D.

In the process state of FIG. **1**D, the contact areas **120**, **122** are buried or embedded in the second structure **136**. Electrical vias to the contact areas **120**, **122** will be produced in the next step according to FIG. **1**E. In this process, contact holes **142**, **144** are produced by etching, for example by plasma assisted dry etching (RIE/ICP) whereby the contact areas **120** and **122** are exposed at the bottom of the contact holes **142**, **144**. Etching chemicals are preferentially used which do not attack the contact areas **120**, **122**, and thus, the etching process is self-terminating. The contact holes **142**, **144** may be cleaned thereafter by, e.g., wet-chemical HCl or H_2SO_4 cleaning, or by plasma cleaning with O_2/Ar/NH_3.

Next, according to FIG. **1**F, a layer of protective material **146** is applied as coating onto the second structure top surface **138** then is formed by the top surface of the protective coating **146**. The protective coating **146** also covers the walls of the contact holes **142**, **144**. The protective coating may be based on nitride, and may comprise AlN or SiN, for example. The deposition of the protective coating **146** should be conformal, creating a layer with 5-10 nm thickness. Material deposition may be performed by atomic layer deposition (ALD) or sputtering. The protective coating **146** may absorb residual stress created by the subsequent steps, provide adhesion and inhibit the fusion of material deposited in the subsequent step(s).

Next, a metal film may be applied (not shown) as a galvanic seed layer to provide electrical conductivity for galvanically filling the contact holes **142**, **144** with an electrically conducting material. Then, the contact holes **142**, **144** are filled with the electrically conducting material in a galvanic process. By filling the contact holes **142**, **144** with an electrically conducting material, vias **148**, **150** are created through the planar second structure top surface **38** down to the electrical contact areas **120**, **122**.

Bonding areas **152**, **154** may be provided on the planar second structure top surface **138**, as shown in FIG. **1**G. The second structure top surface **138** forms the uppermost top surface of the VCSEL device **100** in this embodiment. The bonding areas **152**, **154** may be provided with solder balls (not shown) for connecting electrical links to an external driver (not shown). Thus, the VCSEL device **10** is fabricated, wherein the VCSEL device **10** is fully enclosed by electrical isolating and thermally conductive material **128**, **140**, **146**. In particular, the bonding areas **152**, **154** for bonding the VCSEL device **10** to e.g. an external driver, is located directly above the VCSEL device or chip **100**, thus reducing the footprint of the bonding area by a factor of two or more in comparison with prior art VCSEL devices.

FIGS. **2**A-**2**C show another embodiment of a method of fabricating a VCSEL device **200** shown in FIG. **2**C. The embodiment of FIG. **2**A-**2**C is a modification of the embodiment described before. The VCSEL device **200** comprises a first structure **212** comprising a VCSEL layer structure **214** on a wafer **216**. The layer structure **214** comprises two mesas **202**, **204**, each forming a VCSEL.

FIG. **2**A shows a state of the method of fabricating the VCSEL device **200** which corresponds to the state in FIG. **1**C of the method described above, i.e. the processing sequence in FIGS. **1**A and **1**B is omitted here for the sake of simplification. FIG. **2**A thus shows the VCSEL structure (second structure) **236** which has been planarized before as described above so that it has the planar second structure top surface **238**. A working (polishing) stop layer is not shown here, but may be provided as described above.

In the present embodiment, the first structure **212** includes electrical contact areas **220***a*, **220***b*, **220***c* and **220***d*. Starting from the state of the VCSEL structure **236** in FIG. **2**A, a part of the layer of cover material **228**, which is electrically isolating, is removed in a region **256***a* and **256***b* by etching, thus exposing areas of the VCSEL structure **236** which are intended for further processing, while the cover material electrically further isolates and encapsulates the mesas **202**, **204**. The regions **256***a*, **256***b* are then filled or partially filled with electrically conducting material up to the height level of the planar second structure top surface **238** to provide vias **252***b* and **252***d*. The planar second structure top surface **238** forms the uppermost top surface of the VCSEL device **200**. Bonding areas **252***a* and **252***c* may then be arranged on the contact areas **220***a*, **220***c*, and the upper surfaces of the vias **252***b* and **252***d* may be used as bonding areas. Thus, all bonding areas are arranged on top of the VCSEL device **200** in a dense pack.

FIGS. **3**A-**3**D show another embodiment of a method of fabricating a VCSEL device **300** as a modification of the previous embodiments. Like in the previous embodiment, the VCSEL device **300** comprises two mesas **302**, **304**. The first structure **312** comprises the VCSEL layer structure **314** on wafer **316**. The first structure comprises contact areas **320***a*, **320***b*, **320***c*, **320***d*. In the embodiment according to FIGS. **3**A-**3**D, planarization of a VCSEL structure is shown with a layer of metallic cover material, as will be described below.

The description starts with reference to FIG. **3**A in a process state of the method, which is the process state shown in FIG. **2**B. That is, the second structure **336** with the layer of cover material **328** has been planarized before and then etched in regions **356***a* and **356***b* as described above.

According to FIG. **3**B, the VCSEL structure (second structure) **336** is covered with a further layer of cover material **340** with a thickness such that a lowest height level **330** of the cover material top surface **332** is at least equal to or above the highest level of the non-planar top surface of the second structure **336** shown in FIG. **3**A. The second structure **336** and the cover material **340** form a third structure **337**. The cover material **340** here is a metallic material, e.g. Au. Next, the top surface **332** of the third structure is worked, in particular polished, e.g. using chemical mechanical polishing with a slurry. The slurry may include particles inside the polishing liquid suitable for polishing the metallic cover material **340**. As shown in FIG. **3**C, the metal polishing is stopped at the surface of the mesas **302**, **304**, creating an inherent electrical isolation, by the electrically isolating material **328**, between the lower altitude areas in the regions **356***a*, **356***b* and the top of the VCSEL mesas **302**, **304**. The final external metallization or bonding areas **350***a*, **350***b*, **350***c*, **350***d* shown in FIG. **3**D can then be applied on a flat VCSEL device uppermost top surface **335**, while the contact areas **320***b*, **320***d* in the regions **356***a*, **356***b* are filled with the metallic material **340**.

With reference to FIGS. **4**A-**4**J, another embodiment of a method of manufacturing a VCSEL device **400** shown in FIG. **4**J will be described.

According to FIG. **4**A, a first structure **412** comprising a VCSEL layer structure **414** on a wafer **416** is provided. The

VCSEL layer structure **414** including the wafer **416** comprises one or more semiconductor materials. The VCSEL layer structure **414** may comprise distributed Bragg reflectors and an active region with one or more quantum wells. The VCSEL layer structure **414** may have p- and n-regions. The first structure **412** has a non-planar first structure top surface **418** with varying height levels along the non-planar top surface **418**. The non-planar first structure top surface **418** comprises electrical contact areas **420**, **422** arranged on different height levels on the first structure top surface **418**. According to FIG. **4**A, a working stop layer **424** is applied onto the first structure top surface **418** in the regions thereof with the highest height level as shown in FIG. **4**A. A top surface **426** of the working stop layer **424** forms the top surface of the first structure **412** in this case.

As shown in FIG. **4**B, a layer of cover material **428** different from the semiconductor materials of the first structure **412** is applied on the non-planar first structure top surface **418**, **426** with a thickness such that the lowest height level **430** of a cover material top surface **432** is at least equal to or above the highest height level **434** of the top surface **418**, **426**.

As shown in FIG. **4**C, the second structure **436** comprising the first structure **412** and the cover material **428** is planarized by working the cover material top surface **432** to provide the second structure **436** with a planar second structure top surface **438**. Working the cover material top surface **438** is carried out by polishing, in particular chemical mechanical polishing the cover material top surface **438** as described above. According to FIG. **4**D, a layer of cover material **440** is applied on the planarized second structure **436**. As to the materials of the layer **424**, the cover material **428** and the cover material **440**, reference is made to the description of the embodiment in FIGS. **1**A-**1**G.

According to FIG. **4**E, contact holes **442**, **444** are produced directly above the contact areas **420**, **422**. A protective coating **446** is applied onto the planar second structure top surface **438** as described above. Vias **448**, **450** are then produced by filling the contact holes **442**, **444** with an electrically conductive material. In the present embodiment, the electrically conductive material is for example Au which is galvanically filled into the contact holes **442**, **444**.

Up to here, the process sequence shown in FIGS. **4**A-**4**F may be identical to the process sequence shown in FIGS. **1**A-**1**G, and as far as not indicated otherwise, the description of the method according to FIGS. **1**A-**1**G also applies to the method steps shown in FIGS. **4**A-**4**F.

FIG. **4**G-**4**J show a further process sequence which may be carried out starting from FIG. **4**F.

According to FIG. **4**G, the second structure top surface **438** is planarized again by chemical mechanical polishing the electrically conductive materials forming the vias **448**, **450** with a suitable slurry. The polishing does not attack the protective coating **446**.

Next, as shown in FIG. **4**H, a material **451** is deposited on the vias **448**, **450** which has the function of a defusion stop. The material **451** may be an electrically conductive ceramic material, wherein TiN can be advantageously used as the material **451**. Further, as shown in FIG. **4**H, the second structure is covered with an electrically isolating material **453** to provide a third structure **437** which now has the planar top surface **438**. The VCSEL layer structure now is encapsulated.

In FIG. **4**I, contact holes **455**, **457** are etched into the cover material **453** so that the vias **448**, **450** including the diffusion stops **451** are exposed. FIG. **4**J shows the filling of the contact holes **455**, **457** with an electrically conducting material, e.g. copper, to provide vias **459**, **461** in electrical connection with the vias **448**, **450** and thus with the contact areas **420**, **422**. The vias **459**, **461** can be used as bonding areas with sufficient area for applying soldering balls onto the vias **459**, **461**, in order to provide external connection to a driver (not shown). Again, the uppermost top surface **438** of the VCSEL device **400** is planar.

The VCSEL device **400** is a single mesa VCSEL device. The VCSEL device **400** is a bottom emitter, i.e. laser radiation generated by the VCSEL is emitted through the substrate or wafer **16**.

With reference to FIGS. **5**A-**5**D, an embodiment of a method of fabricating a VCSEL device **500** will be described which exemplarily shows that by repeating the planarization sequence multiple times, electrical connection of multiple VCSELs or multiple mesas at different height levels or altitudes can be achieved. In every planarization and metallization sequence, another part of the mesas (VCSELs) may be connected together, while the external connection to the driver is done at a higher altitude.

The VCSEL device **500** shown in FIG. **5**C comprises, as an example, three mesas or VCSELs **570**, **572**, **574** each enabled to emit laser radiation.

FIG. **5**A shows a processing state of the method of fabricating the VCSEL device **500**, in which a first structure **512** comprising a VCSEL layer structure **514** on a wafer **516** and having a non-planar top surface **518** with electrical contact areas **520***a*-**520***e* has already been covered (a stop layer below the cover material may have been applied before (not shown) like in the first embodiment in FIG. **1**A-**1**G) with an electrically isolating cover material **528** and planarized to provide a second structure **536** with a planar second structure top surface **538**. The processing state of the method in FIG. **5**A corresponds to the processing state of the method shown, for example, in FIG. **4**C.

As shown in FIG. **5**B, the method proceeds with applying a further layer of cover material **540**, contact hole etching and contact hole filling with an electrically conducting material to provide vias **548***a*-**548***e* through the second structure top surface **538** down to the contact areas **520***a*-**520***e* directly above the contact areas **520***a*-**520***e*. The state of the method shown in FIG. **5**B is similar to the state of the method shown in FIG. **4**F.

Next, as shown in FIG. **5**C, a further layer of cover material **553** is applied onto the structure shown in FIG. **5**B to provide a third structure **537** having a third structure top surface **532** forming the uppermost top surface of the fabricated VCSEL device **500**, and contact hole etching is repeated as well as contact hole filling with an electrically conductive material. On top of the third structure top surface **532**, bonding areas or pads **580**, **582**, **584** are produced. The bonding pad **580** is electrically connected to and electrically connects the electrical contact areas **520***a*, **520***c* and **520***d* in parallel, while contact pad **582** is in electrical connection with electrical contact area **520***b*, and bonding pad **584** is electrically connected with contact area **520***e* (see also FIG. **5**D). This exemplary embodiment shows that the method according to the principles of the present disclosure allows for fabricating densely packed VCSEL arrays with highly reduced footprint of the bonding areas.

The method according to the principles described herein also allows for making electrical connections between a plurality of mesas or VCSELs at deeper altitudes, i.e. to make internal connections within the VCSEL device. This will be described hereinafter, first with reference to FIGS. **6**A-**6**H.

FIG. **6**G shows a VCSEL device **600**. The VCSEL device **600** comprises a first structure **612** comprising a VCSEL layer structure **614** comprising two mesas **670**, **672**, on a wafer **616**. The first structure **612** comprises electrical contact areas **620***a*-**620***d*. Within the VCSEL device **600**, an internal electrical link **669** electrically connects the electrical contact areas **620***b*, **620***d* arranged at a low altitude in the VCSEL device **600**, in parallel. In the following, an embodiment of a method to fabricate the VCSEL device **600** will be described with reference to FIG. **6**A-**6**F.

The description starts with a processing state of the method which corresponds to the processing state in FIG. **2**A described above. That is, a planarized second structure **636** has already been formed before, comprising a first structure **612** comprising a VCSEL layer structure **614** on a wafer **616**. Contact areas **620***a*-**620***c* are arranged on the first structure **612**. The first structure **612** has a non-planar first structure top surface **618**, while the second structure **636** has been planarized by working, in particular by chemical mechanical polishing, a cover material **628** applied onto the first structure **612**. The cover material **628** is electrically isolating.

As shown in FIG. **6**B, the second structure **636** is etched so that the deeper lying electrical contact areas **620***b* and **620***d* are exposed, while a part of the remaining cover material **628** electrically isolates the mesas **670**, **672** of the VCSEL layer structure **614**.

The second structure **636** is covered with a further layer of cover material **640** which is an electrically isolating material, as shown in FIG. **6**C.

As shown in FIG. **6**D, the second structure **636** is etched to expose the electrical contact areas **620***b* and **620***d* again, while the mesas **670**, **672** are electrically isolated from the contact areas **620***b* and **620***d* also by the cover material **640**. As shown in FIG. **6**E, the second structure **636** is covered with a metallic material or metallization **641** which electrically connects the electrical contact area **620***b* with the electrical contact area **620***d* by forming electrical vias **648**, **650** to the electrical contact areas **620***b* and **620***d*.

As shown in FIG. **6**F, a further layer of cover material **643** is applied on top of the metallization **641**, to create a third structure **637**, comprising the second structure **636** and the first structure **612**, as shown in FIG. **6**F. The further cover material **643** is an electrically isolating material, and may be a material like the electrically isolating cover materials described with reference to FIG. **1**A-**1**G above.

The third structure **637** has a third structure top surface **639** which is planar and forms the uppermost top surface of the fabricated VCSEL device **600**, wherein the top surface **639** may have been planarized by chemical mechanical polishing.

Next, as shown in FIG. **6**G, the third structure **637** is etched to provide a contact hole **642** which is then filled with an electrically conductive material, which may be the same as the material used for the metallization **641**. The metallization **641** may be Au.

Finally, a bonding area or pad **680** is arranged on the via **651**.

FIG. **6**H shows a top view of the VCSEL device **600** which illustrates the bonding area **680** as well as two further bonding areas **682**, **684**, wherein the bonding area **682** is arranged for electrical connection with the electrical contact area **620***a* and the bonding area **684** is arranged for electrical connection with the electrical contact area **620***c*.

Thus, it has been shown that internal electrical links like link **669** can be made with the method according to the principles of the present disclosure.

FIG. **7** shows an embodiment of a planarized VCSEL device **700** fabricated according to the principles of the present disclosure. The VCSEL device **700** is an example in which a photonic component like a photodiode is integrated in the planarized VCSEL device **700**.

The VCSEL device **700** comprises a first structure **712** comprising a wafer **716** and formed thereon a VCSEL layer structure **714**. The first structure **712** also comprises a photodiode layer structure **715**. The first structure **712** has a first structure top surface **718** which is non-planar as in the embodiments described above. The first structure top surface **718** is also formed in part by a surface of the photodiode layer structure **715**. The first structure **712** also comprises electrical contact areas **720**, **721**, **722**, which are arranged at different height levels above the wafer **716**. The VCSEL device **700** is encapsulated by a layer of electrically isolating cover material **728** made as described above. The layer of cover material **728** together with the first structure **712** forms a second structure **736** having a planarized second structure top surface **738** which forms the uppermost top surface of the VCSEL device **700**. The second structure **736** has been fabricated as described above. Also shown in FIG. **7** is a working (polishing) stop layer **724** applied before applying the layer of cover material **728**. A protective coating **746** as well as electrical vias **748**, **749**, **750** have been produced as described above. Fusion stops **751** on the top surfaces of the respective vias **748**, **749**, **750** are also shown in FIG. **7**.

Bonding areas **759** are arranged on the second structure top surface **738** in electrical connection with the vias **748**, **749**, **750**. Solder balls **780** are arranged on the bonding areas **759** for connection to an external driver (not shown).

The VCSEL device **700** is a bottom emitter, i.e. laser light is emitted by the active region of the VCSEL layer structure **714** through the photodiode layer structure **715** as indicated with an arrow **785**. A grating **790** or other optical structures may be arranged at the light-emitting side of the VCSEL device **700**.

FIG. **7** shows that the method of fabricating a VCSEL device according to the principles of the present disclosure also is suitable for integrating photonic components like a photodiode or phototransistor in the VCSEL device **700**, since all electrical connections to external devices can be made on a planarized surface (surface **738**) of the VCSEL device **700** in a space-saving manner.

While the VCSEL **700** in FIG. **7** is a single-mesa VCSEL device, FIGS. **8**A and **8**B show a multi-mesa VCSEL device **800**, wherein each mesa may be equipped with an integrated photodiode. The VCSEL **800** comprises in this example a first structure **812** comprising a VCSEL layer structure **814** on a wafer **816** including a photodiode layer structure **815**. The first structure **812** has been etched to obtain four mesas **801**, **802**, **803**, **804** and a corresponding number of photodiodes **805**, **806**, **807**, **808**. Each of the mesas **801**, **802**, **803**, **804** including the corresponding photodiode **805**, **806**, **807**, **808** resembles the VCSEL device **700** in FIG. **7**. However, differently from the VCSEL device **700**, the VCSEL device **800** comprises an internal electrical link **831**. Such an internal electrical link is advantageously possible by the method of fabricating the VCSEL device **800** in accordance with the principles of the present disclosure. Such an internal electrical link may be produced by repeating the planarizing sequence including applying layers of cover material and planarizing them several times as described, for example, with reference to FIG. **6**A-**6**H above. After a first planarization step, first electrical vias **848***a* to **848***d* down to the electrical contact areas on the layer structure **814** of the mesas **801**-**804** are produced and electrically connected to one another by applying an electrically conductive material, for example a metal, in order to create the electrical link **831**. Then, after the next planarization process including covering the second structure **836** with a further layer of cover material to provide a third structure **837** and planarizing the third structure **837** to provide a planar uppermost top surface of the device **800**, further electrical vias **870***a*-**870***e* are produced in the third structure **837**. As shown in FIG. **8**A, one of these further vias **870***a* extends down to the electrical link **831** while electrical connection of the via **870***a* to the vias **848***a*, **848***b*, **848***c*, **848***d* is accomplished by the electrical link **831**. Further electrical vias **870***b*-**870***e* have been produced which extend from the third structure top surface **839** down to the electrical contacts arranged on the photodiodes **805**-**808**. A further electrical via **870***f* extends from the third structure top surface **839** down to the wafer **816**. Thus, the method of fabricating a VCSEL device like the VCSEL device **800** enables densely packed multi-mesa VCSEL devices, whereby dark spots in the laser emission are avoided or at least reduced. Further, as shown in FIG. **8**A, the method according to embodiments of the present invention allows providing internal electrical links within a VCSEL device, like the internal electrical link **831**.

As shown in the top view of the VCSEL device **800** in FIG. **8**B, the bonding areas including the solder balls **880***a*-**880***f* are all arranged on the top surface of the VCSEL device **800** without requiring much space reducing the footprint of the VCSEL device **800**, in particular the footprint of the electrical connections.

FIG. **9** shows an embodiment of a cascaded VCSEL device **900** fabricated in accordance with the principles of the present disclosure, in which multiple mesas or VCSELs are electrically connected in series with one another. The VCSEL device **900** comprises a first structure **912** comprising a VCSEL layer structure **914**, which has a region **917** of a first polarization type, for example p-type, and another region with a second polarization type, for example an n-type region **919**. The VCSEL device **900** comprises a second structure **936**, which is formed from the first structure **912** by a planarizing process as described above. The first structure **912** also comprises a wafer **916**.

The VCSEL device **9** further comprises a third structure comprising a further layer of cover material **953**.

The VCSEL device **900** comprises in this example four mesas **901**-**904**. Each mesa has a structure comparable with the VCSEL structure shown in FIG. **4**J described above. Vias **959***a*-**959***e* are produced in the cover material layer **953**. The vias **959***b*, **959***c* and **959***d* electrically connect adjacent mesas with one another. The whole arrangement is made such that the mesas are electrically connected in series, so that a cascade of mesas is formed. When connecting the bonding area **980***a* formed as a solder ball and **980***e* with the positive and negative poles of a voltage source, the voltage (for example 12V) drops along the VCSEL device **900**. The current flows through the single mesas **901**, **902**, **903**, **904** in series, as indicated with a broken partial line **990**.

The further solder balls **980***b*, **980***c*, **980***d* as well as the solder balls **980***a*, **980***e* also serve as heat sinks, so that each emitter (mesa) has its own heat sink.

FIG. **10** shows an embodiment of a VCSEL device **1000** which is arranged for laser light emission in a plurality of different wavelength bands, for example in a first wavelength band with a peak emission at 980 nm, a second wavelength band with a peak emission at 960 nm, and a third wavelength band with a peak emission at 940 nm.

The regions of the VCSEL device **1000** with the emissions in the different wavelength bands are indicated by arrows **1001**, **1002**, **1003**. In particular, the VCSEL device **1000** is a top emitter.

The VCSEL **1000** may be fabricated in accordance with the principles of the present disclosure, i.e. by providing a first structure comprising a VCSEL layer structure **1014** with regions **1014***a*, **1014***b*, **1014***c* in accordance with the number of emission peak wavelengths of the laser emission. The VCSEL layer structure **1014** comprises, for example, an n-type doped layer **1017** common to all regions **1014***a*, **1014***b*, **1014***c*. The first structure **1012** further comprises a wafer or substrate **1016**. The VCSEL device **1000** may fabricated in accordance with the principles of the present disclosure by using a planarizing sequence as described above, and producing electrical vias from the structure top surface down to the electrical contact areas as shown in FIG. **10**. The VCSEL device **1000** has bonding areas **1070**, **1071**, **1072**, **1073** for electrically connecting the VCSEL device **1000** to an external driver, wherein the regions **1001**, **1002**, **1003** are individually addressable.

As it becomes apparent from the description herein, the method of fabricating a VCSEL device according to embodiments of the present invention enables a multitude of different configurations of VCSEL devices with low manufacturing expenditure. In particular, the method allows for VCSEL devices with a small footprint of the VCSEL chip. The method allows for fabricating VCSEL devices with individually addressable VCSELs or mesas in a highly dense pack, allows photonic component integration, and complex internal electrical routing.

While subject matter of the present disclosure has been illustrated and described in detail in the drawings and foregoing description, such illustration and description are to be considered illustrative or exemplary and not restrictive. Any statement made herein characterizing the invention is also to be considered illustrative or exemplary and not restrictive as the invention is defined by the claims. It will be understood that changes and modifications may be made, by those of ordinary skill in the art, within the scope of the following claims, which may include any combination of features from different embodiments described above.

The terms used in the claims should be construed to have the broadest reasonable interpretation consistent with the foregoing description. For example, the use of the article "a" or "the" in introducing an element should not be interpreted as being exclusive of a plurality of elements. Likewise, the recitation of "or" should be interpreted as being inclusive, such that the recitation of "A or B" is not exclusive of "A and B," unless it is clear from the context or the foregoing description that only one of A and B is intended. Further, the recitation of "at least one of A, B and C" should be interpreted as one or more of a group of elements consisting of A, B and C, and should not be interpreted as requiring at least one of each of the listed elements A, B and C, regardless of whether A, B and C are related as categories or otherwise. Moreover, the recitation of "A, B and/or C" or "at least one of A, B or C" should be interpreted as including any singular entity from the listed elements, e.g., A, any subset from the listed elements, e.g., A and B, or the entire list of elements A, B and C.

The invention claimed is:

1. A method of fabricating a Vertical Cavity Surface Emitting Laser (VCSEL) device, the method comprising:

providing a first structure comprising a VCSEL layer structure on a wafer, the first structure including the wafer comprising one or more semiconductor materials, the first structure having a non-planar first structure top surface with varying height levels along the non-planar top surface, wherein the non-planar first structure top surface comprises one or more electrical contact areas at different height levels above the wafer;

applying one or more layers of a cover material that is different from the one or more semiconductor materials on the non-planar first structure top surface with a thickness such that a lowest height level of a cover material top surface is equal to or above the highest height level of the non-planar first structure top surface, to obtain a second structure comprising the first structure and the one or more layers of cover material, the second structure having a second structure top surface;

planarizing the second structure top surface; and

producing one or more first electrical vias from the second structure top surface through the one or more layers of cover material for electrical connection to the one or more electrical contact areas.

2. The method of claim **1**, wherein the planarizing comprises polishing the second structure top surface.

3. The method of claim **2**, wherein the polishing comprises chemical mechanical polishing.

4. The method of claim **1**, wherein at least one of the one or more layers of cover material is electrically isolating.

5. The method of claim **1**, wherein at least one of the one or more layers of cover material is a metal layer.

6. The method of claim **1**, wherein at least one of the one or more layers of cover material is thermally conducting.

7. The method of claim **1**, wherein the producing the one or more first electrical vias comprises etching one or more contact holes into the one or more layers of cover material down to the one or more electrical contact areas, and filling the one or more contact holes with an electrically conducting material up to the second structure top surface.

8. The method of claim **1**, further comprising, after producing the one or more first electrical vias, applying at least one further layer of cover material on the second structure to provide a third structure comprising the second structure and the further cover material and having a third structure top surface.

9. The method of claim **8**, further comprising producing further electrical vias from the third structure top surface through the at least one further layer of cover material for electrically connecting the one or more further electrical vias with at least a part of the first electrical vias.

10. The method of claim **8**, further comprising, prior to applying the at least one further layer of cover material, electrically connecting at least a part of the first electrical vias with one another.

11. The method of claim **1**, further comprising arranging one or more bonding areas on the second structure top surface or on the third structure top surface in electrical connection with the first or further electrical vias.

12. The method of claim **1**, wherein the one or more semiconductor materials are II-VI or III-V compound semiconductor materials.

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